

Simplex IV: Column Geometry and Efficiency

How does a pivot change the picture, why can simplex be slow, and what other methods solve LPs?

Part A · From a bounded LP to mixture weights

3.6 · Column geometry and a complete pivot

Part B · Efficiency and the Hirsch conjecture

LP algorithms · History and alternatives

Lecture 8 · Bertsimas–Tsitsiklis §§3.6–3.7

Build on feasible bases, reduced costs, the ratio test, and the two-phase method from Simplex I–III.

Optional study: proofs, a second example, implementation, and complexity

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The same LP, viewed in two different spaces

The picture we already know

A feasible solution $x = (x_1, \dots, x_n)$ is one point in **variable space**. A simplex step moves along an edge, unless it is degenerate.

A picture built from the columns

Each A_j is a point in **constraint-coordinate space**; x_j supplies its weight.

$$Ax = \sum_j x_j A_j = b.$$

With two coordinate equations, we can draw the columns on a floor and add cost as height. More coordinate equations give the same idea in higher dimensions.

These are different spaces: the column picture is not a projection of the original feasible region.

Why do the weights need to sum to one?

In column space, feasibility is a combination of the columns:

$$Ax = \sum_j x_j A_j = b, \quad x_j \geq 0.$$

If the weights also satisfy $\sum_j x_j = 1$, the combination is a **weighted average**. It lies in the **convex hull** of the column-points.

Adding $\sum_j x_j = 1$ to an existing LP can exclude feasible points.

Our question: can we obtain weights summing to one by changing variables and preserving the original problem?

Start with an LP whose variables do not sum to one

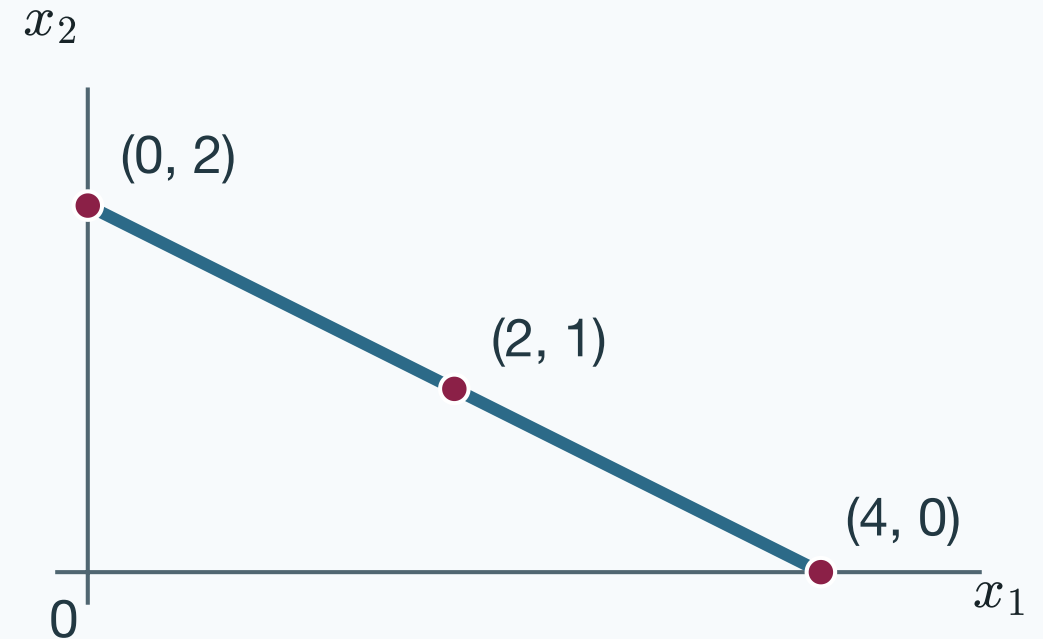
$$\min f = -3x_1 - x_2 \quad \text{s.t.} \quad x_1 + 2x_2 = 4, \quad x_1, x_2 \geq 0.$$

The equality and nonnegativity give a bounded feasible segment.

At $(4, 0)$, the variables sum to 4. At $(0, 2)$, they sum to 2.

They cannot serve directly as weights summing to one.

We will represent every point on this segment using new weights, while keeping its objective value.



Every feasible point lies on this segment.

Add the unused part of a common bound

For every feasible point of $x_1 + 2x_2 = 4$, $x_1, x_2 \geq 0$,

$$x_1 + x_2 = 4 - x_2 \leq 4.$$

Define the unused amount s :

$$s = 4 - x_1 - x_2 \geq 0 \quad \Longleftrightarrow \quad x_1 + x_2 + s = 4.$$

No original point is removed. Every feasible (x_1, x_2) determines exactly one nonnegative s .

Now all three nonnegative quantities have the same total: 4.

Divide by the common total to get weights

$$\lambda_1 = \frac{x_1}{4}, \quad \lambda_2 = \frac{x_2}{4}, \quad \lambda_0 = \frac{s}{4}.$$

The subscript 0 identifies the extra weight for the unused amount.

$$\begin{array}{ll} \min & -12\lambda_1 - 4\lambda_2 \\ \text{s.t.} & \lambda_1 + 2\lambda_2 = 1, \\ & \lambda_0 + \lambda_1 + \lambda_2 = 1, \\ & \lambda_0, \lambda_1, \lambda_2 \geq 0. \end{array}$$

Recover the original variables with $x_1 = 4\lambda_1$, $x_2 = 4\lambda_2$; hence $-3x_1 - x_2 = -12\lambda_1 - 4\lambda_2$.

Corresponding points have the same cost

Use $s = 4 - x_1 - x_2$ and $(\lambda_0, \lambda_1, \lambda_2) = (s, x_1, x_2)/4$.

Original point (x_1, x_2)	Unused amount s	New weights $(\lambda_0, \lambda_1, \lambda_2)$	Cost in both models
$(4, 0)$	0	$(0, 1, 0)$	-12
$(2, 1)$	1	$(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$	-7
$(0, 2)$	2	$(\frac{1}{2}, 0, \frac{1}{2})$	-2

Every row has nonnegative weights summing to 1, and $\lambda_1 + 2\lambda_2 = 1$.

These are three examples. Next we prove that the construction works for **every** feasible point of any bounded standard-form LP.

Boundedness gives one bound for all feasible points

Consider $\min c^\top x$ over a nonempty, bounded set

$$P = \{x : Ax = b, x \geq 0\}.$$

Choose one number $M > 0$ such that

$$\sum_{j=1}^n x_j \leq M \quad \text{for every } x \in P.$$

Why does one exist? Boundedness gives finite coordinate bounds $x_j \leq U_j$. Any positive $M \geq \sum_j U_j$ works.

The bound need not be tight. If all $U_j = 0$, choose any positive M .

The complete normalized LP

Let $s = M - \mathbf{1}^\top x$, $\lambda = x/M$, and $\lambda_0 = s/M$. Here $\mathbf{1}$ is the vector of ones.

$$\begin{array}{ll}\min & Mc^\top \lambda \\ \text{s.t.} & A\lambda = b/M, \\ & \mathbf{1}^\top \lambda + \lambda_0 = 1, \\ & \lambda \geq 0, \quad \lambda_0 \geq 0.\end{array}$$

The extra variable has zero coefficients in $A\lambda = b/M$ and zero objective coefficient.

We must check both directions: original feasible points give feasible weights, and feasible weights recover original feasible points.

From an original feasible point to feasible weights

Take any $x \in P$. Define $\lambda = x/M$, $\lambda_0 = (M - \mathbf{1}^\top x)/M$.

Nonnegativity: $x \geq 0$, $M > 0$, and $\mathbf{1}^\top x \leq M$ give $\lambda \geq 0$ and $\lambda_0 \geq 0$.

Original equations, scaled:

$$A\lambda = A(x/M) = (Ax)/M = b/M.$$

Weights sum to one:

$$\mathbf{1}^\top \lambda + \lambda_0 = \frac{\mathbf{1}^\top x}{M} + \frac{M - \mathbf{1}^\top x}{M} = 1.$$

From feasible weights back to an original point

Take any feasible (λ, λ_0) of the normalized LP. Set $x = M\lambda$.

Nonnegativity: $M > 0$ and $\lambda \geq 0$ give $x \geq 0$.

Original equations:

$$Ax = A(M\lambda) = M(A\lambda) = M(b/M) = b.$$

The extra weight is exactly the unused amount:

$$M\lambda_0 = M(1 - \mathbf{1}^\top \lambda) = M - \mathbf{1}^\top x = s.$$

Thus $x \in P$, and the two transformations undo each other.

The objective value is preserved as well

Corresponding feasible solutions satisfy $x = M\lambda$.

$$\underbrace{c^\top x}_{\text{original cost}} = c^\top (M\lambda) = \underbrace{Mc^\top}_{\text{normalized cost}} \lambda.$$

Every feasible point has exactly one corresponding set of weights, with the same cost.

Therefore an optimal solution of either model gives an optimal solution of the other.

A better original point would give better normalized weights, and vice versa.

What changed: the coordinates and one extra variable. **What stayed:** the feasible choices and their objective values.

A finite optimum does not make the feasible set bounded

The construction requires a bound on **every feasible point**.

$$\min x_1 \quad \text{s.t.} \quad x_1 - x_2 = 0, \quad x_1, x_2 \geq 0.$$

The optimum is 0 at $(0, 0)$, but (t, t) is feasible for every $t \geq 0$. No finite M bounds $x_1 + x_2 = 2t$.

Simplex itself needs no sum-to-one condition. This reformulation gives a convex-hull picture when the feasible set is bounded.

Without normalization, nonnegative combinations of lifted columns (A_j, c_j) form a **cone**; coefficients need not sum to one.

Why the next column picture fits in three dimensions

For a normalized model with two coordinate equations, each full constraint column is

$$\tilde{A}_j = \begin{pmatrix} a_{1j} \\ a_{2j} \\ 1 \end{pmatrix}.$$

The last coordinate is always 1. Draw the floor point $A_j = (a_{1j}, a_{2j})$; the sum-to-one row ensures **convex weights**.

Add the objective coefficient as height:

$$P_j = (a_{1j}, a_{2j}, c_j).$$

The next example calls its mixture weights x_j again. Each x_j is a weight, while each P_j is a plotted column-point.

Write the mixture LP one equation at a time

For this new example, choose five mixture weights. Two coordinate equations fix the target; the third makes the weights sum to one:

$$\begin{aligned} \min \quad & z = 7x_B + 4x_C + 4x_D + x_E + 4x_F \\ \text{s.t.} \quad & \frac{1}{2}x_B + 3x_C + 2x_E = 1, \\ & \frac{1}{2}x_B + 3x_D + \frac{1}{2}x_E = 1, \\ & x_B + x_C + x_D + x_E + x_F = 1, \\ & x_B, x_C, x_D, x_E, x_F \geq 0. \end{aligned}$$

The first two equalities locate the weighted average on the floor. The sum-to-one equality makes it a convex combination. This example is already in mixture form. Its variables are the weights, not the original variables before normalization.

Add cost as the height of each column

Start with column C: $A_C = (3, 0)^\top$. Its two equation coefficients locate it on the floor.

Attach its unit cost $c_C = 4$ as a third coordinate:

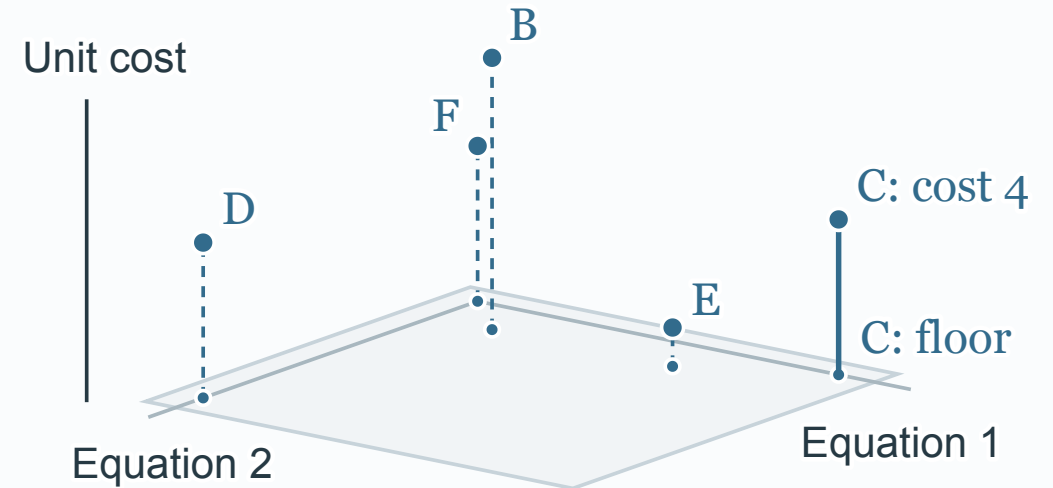
$$P_C = (3, 0, 4).$$

Do the same for all five columns:

$$P_j = \begin{pmatrix} A_j \\ c_j \end{pmatrix}.$$

The vertical segments show the attached costs.

The variables x_j will tell us how to mix these points.



A feasible mixture lands above the target

Give C, D, and F equal weights:

$$x_C = x_D = x_F = \frac{1}{3}, \quad x_B = x_E = 0.$$

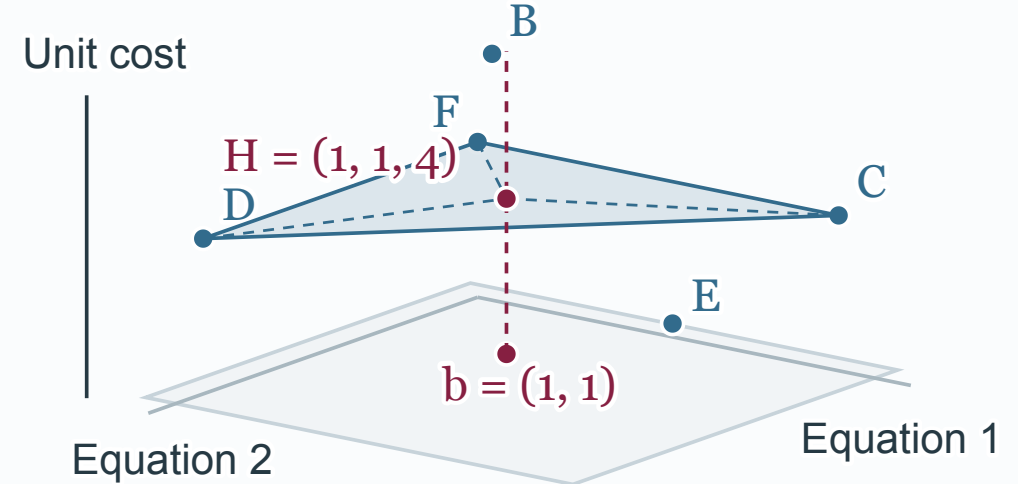
Their lifted points are $P_C = (3, 0, 4)$, $P_D = (0, 3, 4)$, and $P_F = (0, 0, 4)$.

$$\begin{aligned} H &= \frac{1}{3}P_C + \frac{1}{3}P_D + \frac{1}{3}P_F \\ &= (1, 1, 4). \end{aligned}$$

Horizontal coordinates: $b = (1, 1)$.

Height: objective value $z = 4$.

In general, the same weights give $\sum_j x_j P_j = (Ax, c^\top x) = (b, z)$.



Find the lowest attainable cost above the target

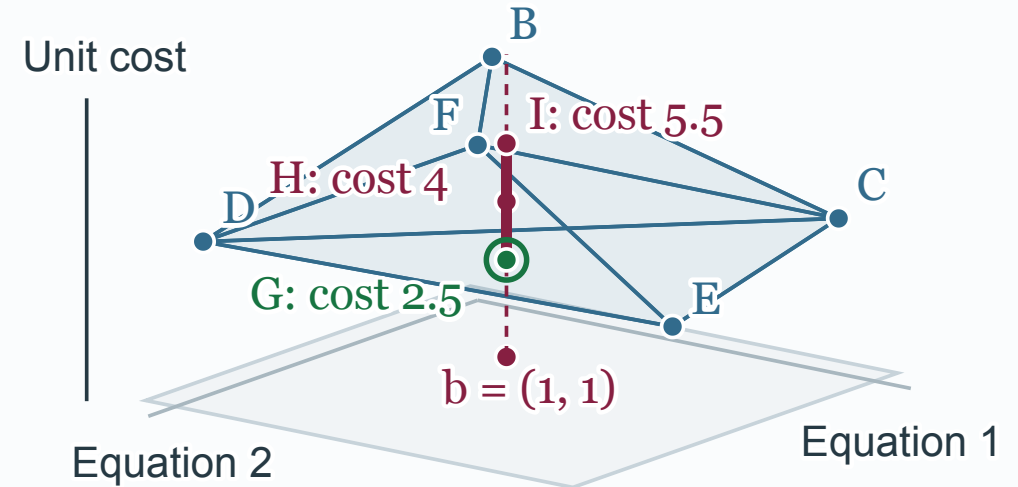
All nonnegative weighted averages with weights summing to one fill the convex hull of B–F.

The **requirement line** contains all $(1, 1, z)$: the two requirements stay fixed while cost varies.

Only the part of that line inside the hull is attainable:

$$\frac{5}{2} \leq z \leq \frac{11}{2}.$$

Minimization chooses the lowest point, G, with $z = 5/2$. H has cost 4; I has cost $11/2$.



Find the three basic weights by substitution

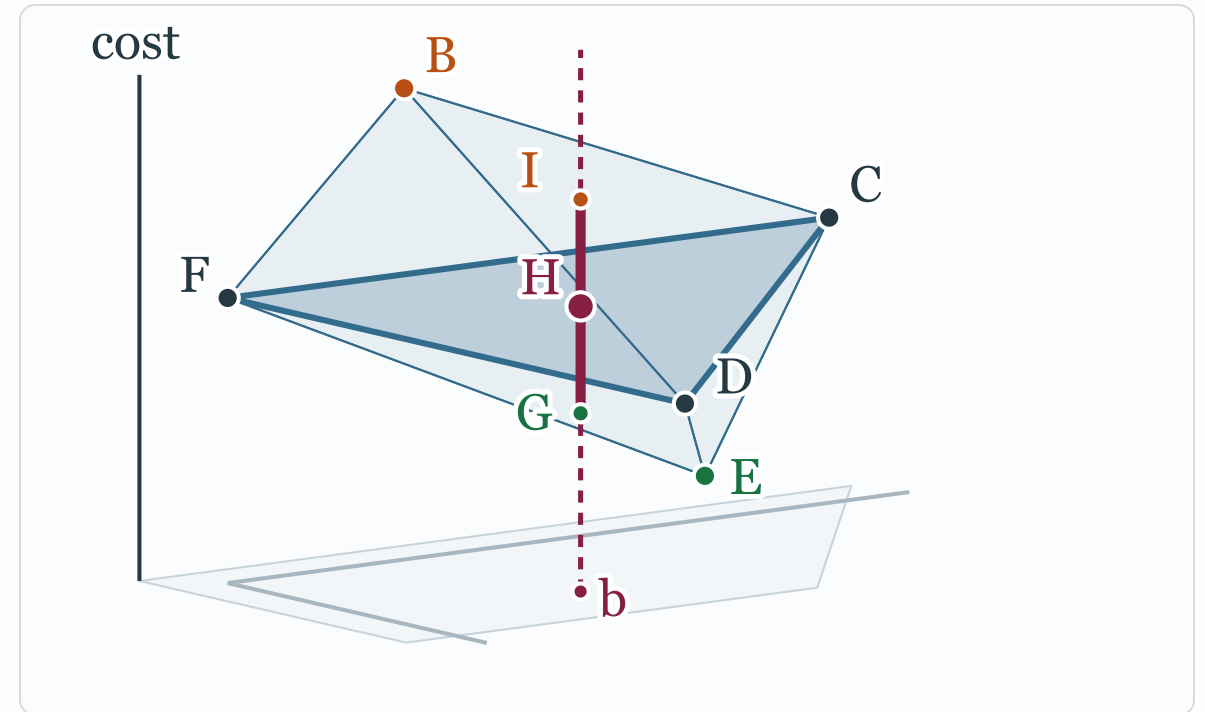
Choose C, D, F as basic. Set the two other weights to zero: $x_B = x_E = 0$.

$$\begin{aligned} 3x_C &= 1, \\ 3x_D &= 1, \\ x_C + x_D + x_F &= 1. \end{aligned}$$

$$x_C = x_D = x_F = \frac{1}{3}.$$

These nonnegative weights give $H = (1, 1, 4)$, the same feasible mixture as before.

The three basic points form triangle CDF.



Keep B and E free; solve for C, D, and F

Solve coordinate equation 1 for C and equation 2 for D. Then substitute both into the sum-to-one equation to obtain F:

$$\begin{aligned}x_C &= \frac{1}{3} - \frac{1}{6}x_B - \frac{2}{3}x_E, \\x_D &= \frac{1}{3} - \frac{1}{6}x_B - \frac{1}{6}x_E, \\x_F &= 1 - x_B - x_E - x_C - x_D \\&= \frac{1}{3} - \frac{2}{3}x_B - \frac{1}{6}x_E.\end{aligned}$$

At $x_B = x_E = 0$, this returns the basic weights $(1/3, 1/3, 1/3)$. Increasing either nonbasic weight changes all three basic weights.

Substitute into the objective to read its slopes

The three basic points C, D, F all have cost 4. Use $x_C + x_D + x_F = 1 - x_B - x_E$:

$$\begin{aligned} z &= 7x_B + 4(x_C + x_D + x_F) + x_E \\ &= 7x_B + 4(1 - x_B - x_E) + x_E \\ &= 4 + 3x_B - 3x_E. \end{aligned}$$

The objective coefficients are the reduced costs: $\bar{c}_B = 3$ and $\bar{c}_E = -3$.

For minimization, increasing E improves the objective if a positive feasible step is available. Increasing B makes it worse.

How far can E increase before a basic weight reaches zero?

Let $x_E = \theta$, starting from $x_C = x_D = x_F = 1/3$.
Preserving all three equalities gives

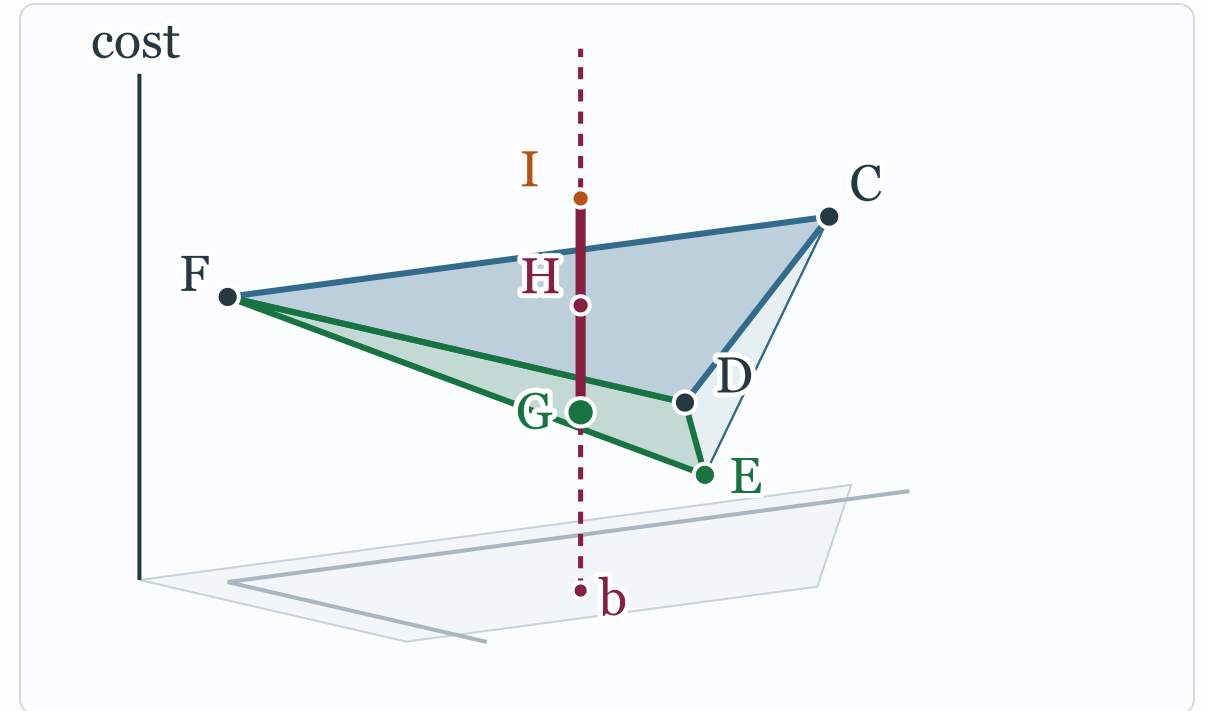
$$x_C = \frac{1}{3} - \frac{2}{3}\theta,$$

$$x_D = \frac{1}{3} - \frac{1}{6}\theta,$$

$$x_F = \frac{1}{3} - \frac{1}{6}\theta.$$

$$\theta^* = \min\{1/2, 2, 2\} = 1/2.$$

The three bounds come from C, D, F respectively. C reaches zero first, so C leaves.



One pivot lowers the attainable cost: H to G

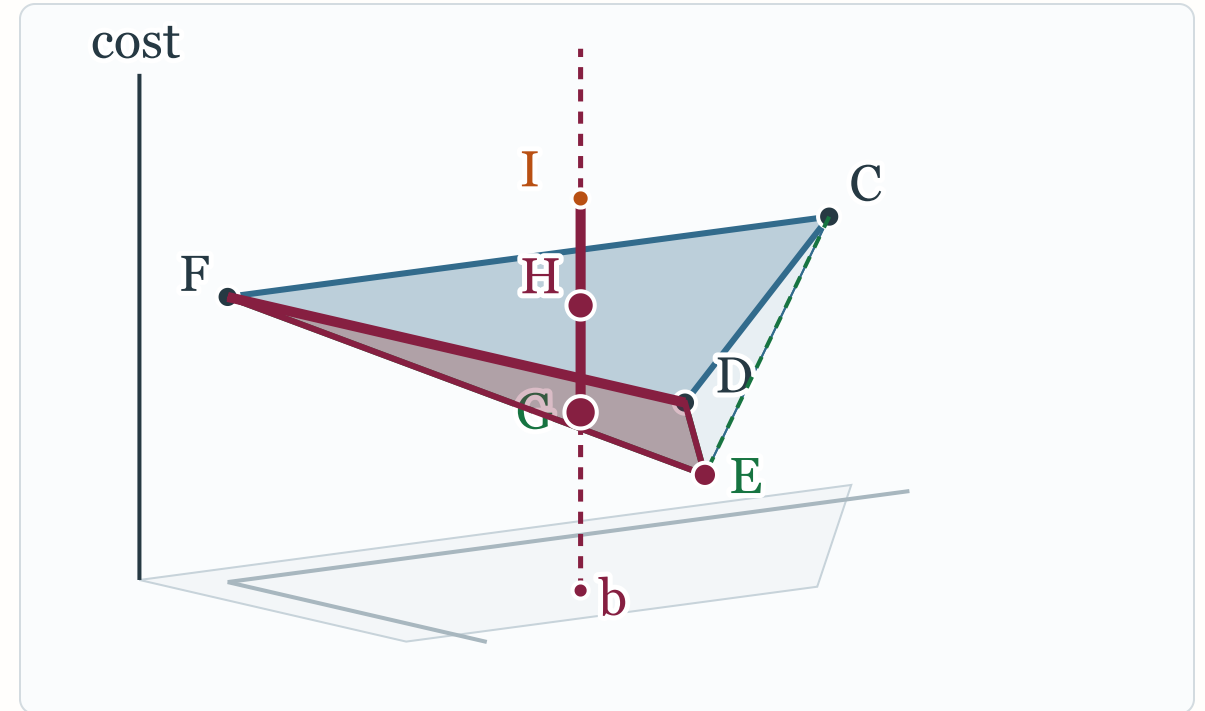
Start with basic columns C, D, F. Their equal-weight mixture is H, with cost 4.

E enters; C leaves. The new basic triangle DEF meets the same requirement line at G.

$$x_D = \frac{1}{4}, \quad x_E = \frac{1}{2}, \quad x_F = \frac{1}{4}, \quad z = \frac{5}{2}.$$

A positive improving step lowers the cost. A degenerate pivot can change the basis without moving this point.

The moving corner illustrates a hinge; intermediate corner positions are not extra LP columns.



New triangle DEF; cost 2.50.

The final plane supports the entire convex hull

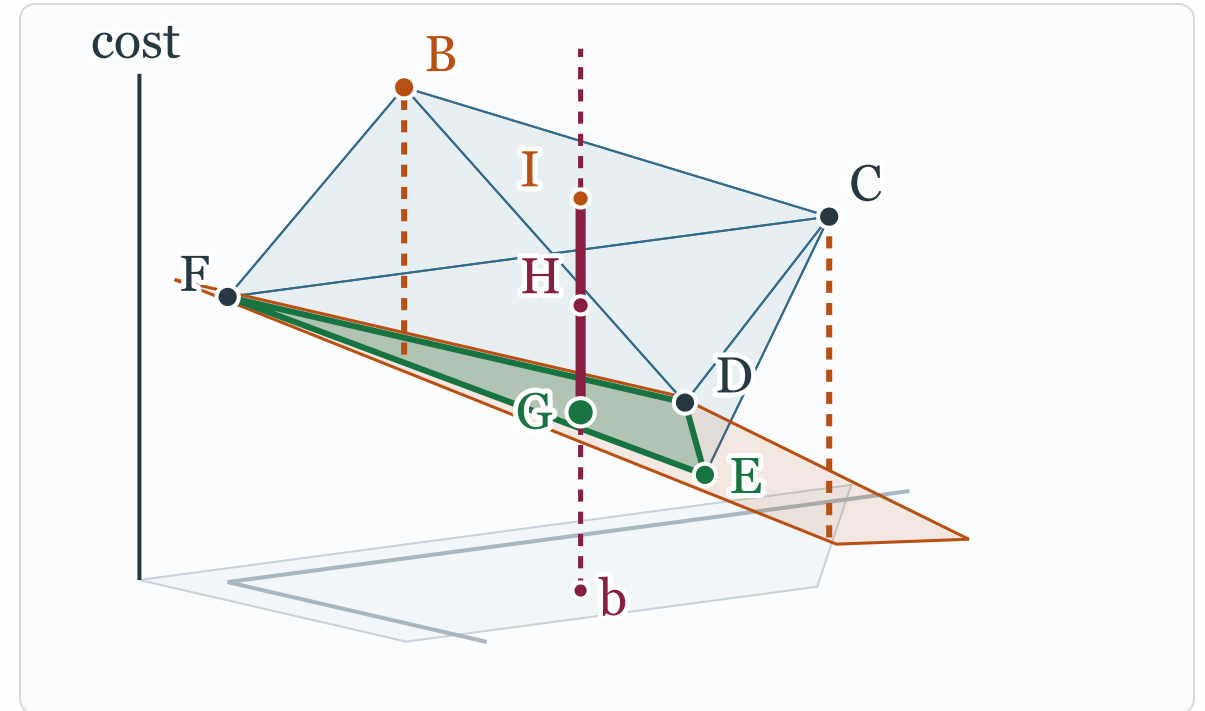
The plane through D, E and F is

$$z = 4 - \frac{3}{2}a_1.$$

Every lifted column lies on or above it. Therefore every convex combination satisfies $z \geq 4 - \frac{3}{2}a_1$.

At $b = (1, 1)$, this gives $z \geq 5/2$. The mixture G attains the bound, so it is optimal.

This is a lower-bound certificate. We will develop that idea in duality.



Termination is only the beginning of the cost question

Lexicographic and Bland rules prevent cycling. They do not promise few pivots.

$$\text{total work} \approx \text{work per pivot} \times \text{number of pivots.}$$

One pivot

With m equations and n variables, row operations update about mn tableau entries: $O(mn)$ arithmetic operations.

The whole run

Even inexpensive steps can add up. We need to understand how many steps the chosen rule takes.

Next: a tilted cube with a long improving path—and a short alternative.

Can an improving path visit every corner?

Choose $0 < \epsilon < \frac{1}{2}$ and solve

$$\begin{array}{ll}\min & -x_n \\ \text{s.t.} & \epsilon \leq x_1 \leq 1, \\ & \epsilon x_{i-1} \leq x_i \leq 1 - \epsilon x_{i-1}, \quad i = 2, \dots, n.\end{array}$$

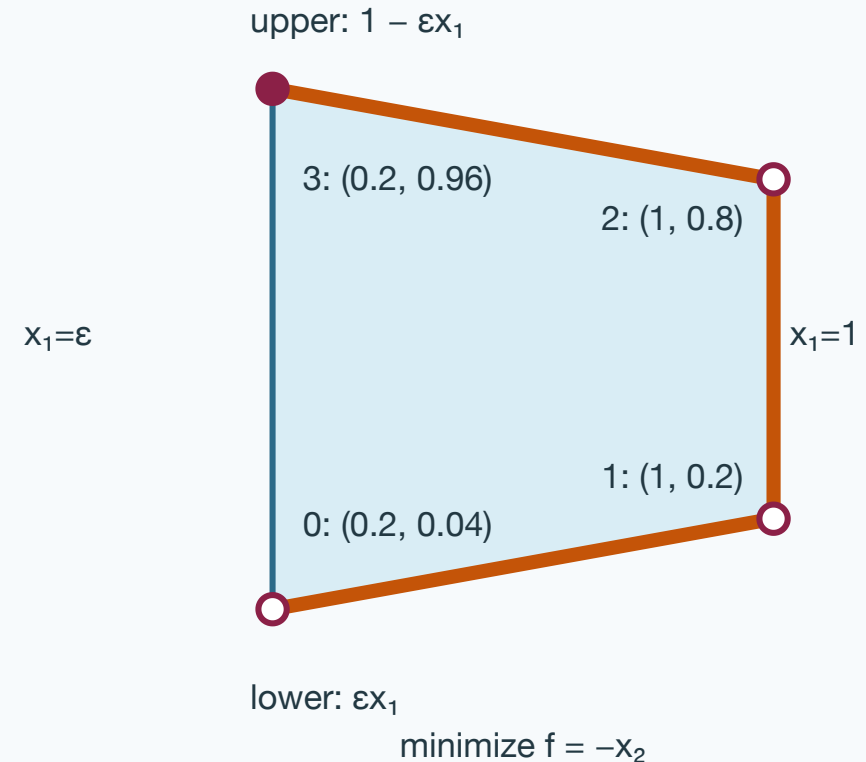
Each coordinate still has a lower and an upper bound. Their dependence on the previous coordinate makes the long path strictly improve the objective.

See the improvement in two dimensions

Set $\epsilon = 0.2$. Start at the vertex labeled 0 and follow the orange path.

vertex	0	1	2	3
x_2	.04	.20	.80	.96
$f = -x_2$	-.04	-.20	-.80	-.96

All three edge moves strictly decrease the objective. The fourth vertex is optimal.

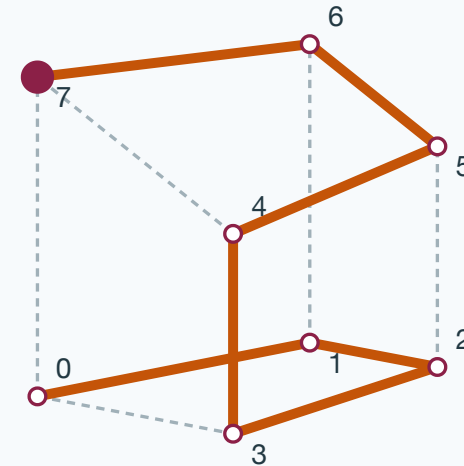


The same construction gives seven improving edges

Use the same inequalities with $n = 3$ and $\epsilon = 0.2$.

The figure shows the complete path from vertex 0 to vertex 7. The objective $f = -x_3$ decreases at every step.

The path visits all 8 vertices. In dimension n , it visits all 2^n vertices.



orange: visited edges · ring: current vertex

Vertex 7 of 7; $x = (0.2, 0.04, 0.992)$; $f = -0.992$.

Theorem 3.5: an exponential pivot path exists

For the perturbed cube just defined:

1. The feasible set has 2^n vertices.
2. They can be ordered so consecutive vertices share an edge and the cost strictly decreases.
3. A pivoting rule that follows this order takes $2^n - 1$ pivots.

This is an existence statement about a bad path and a rule that follows it. It does not say every rule follows that path.

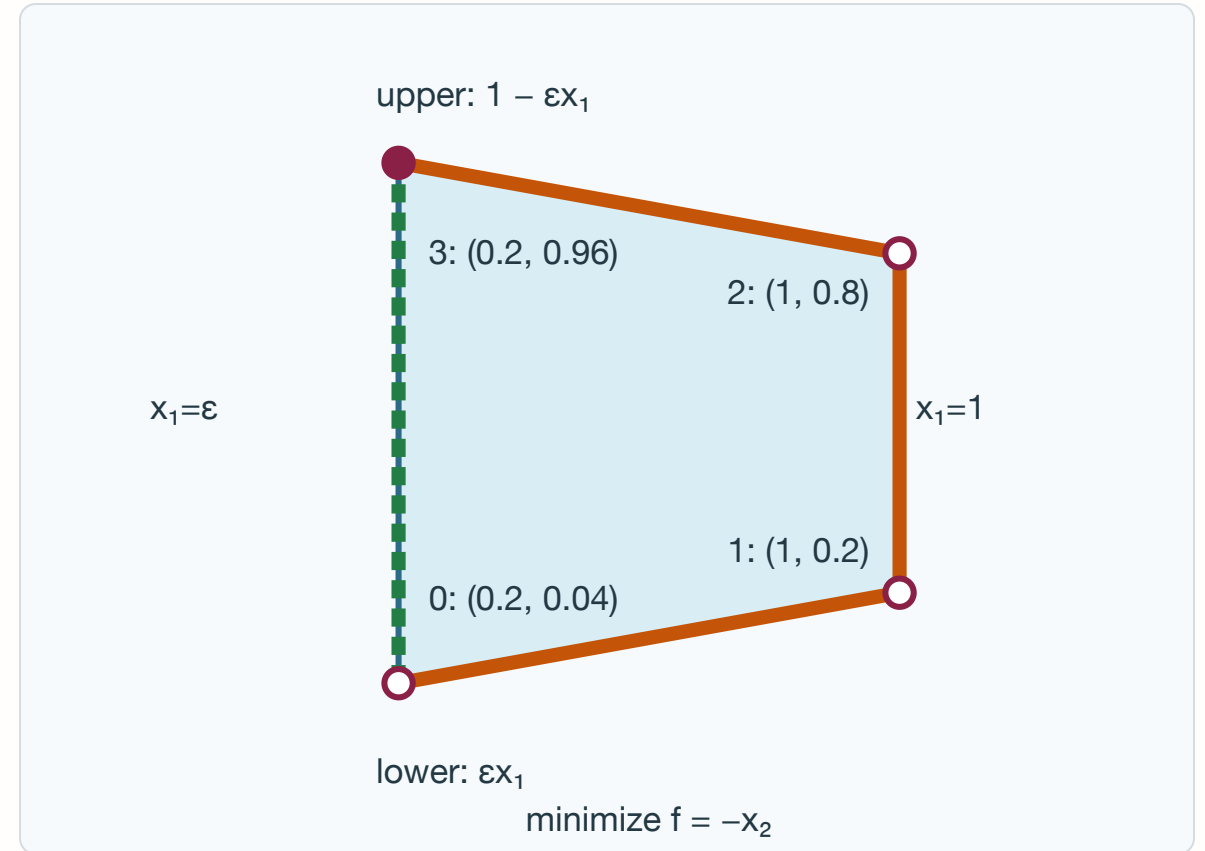
A different pivot choice can take a shortcut

The first and last vertices of this spanning path are adjacent.

Orange: follow the three-edge detour. Green: move directly to the optimum in one pivot.

The same first-to-last adjacency holds in the higher-dimensional construction.

Choosing a rule matters. A long available path does not force every algorithm to use it.



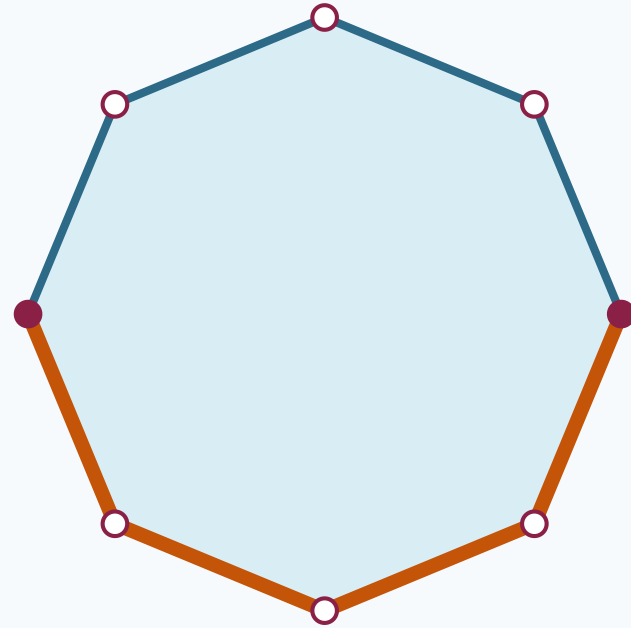
A short available path is different from the path we choose

Join vertices that share an edge. The **graph diameter** is the largest shortest-path distance between two vertices.

For this octagon, the farthest pair is four edges apart. Its graph diameter is 4.

These shortest paths need not improve the objective. A pivot rule may follow a longer route.

How large can this diameter be in higher dimensions?



cycle: $D = 4 = \text{floor}(m / 2) \cdot m = 8$

The Hirsch conjecture is false—even for bounded polytopes

For a bounded polytope of dimension d with F facets, the conjecture proposed

$$\text{graph diameter} \leq F - d.$$

A facet is a boundary face of dimension $d - 1$.

Santos's counterexample

Announced in 2010; published in 2012. A **43-dimensional polytope with 86 facets** has diameter greater than

$$86 - 43 = 43.$$

This disproves the proposed linear bound. It does not rule out polynomial-time LP algorithms or say which path simplex will follow.

Santos, *A counterexample to the Hirsch conjecture*. This later result updates the textbook discussion.

Three milestones in solving linear programs

The long simplex path raises a broader question: can another kind of algorithm have a polynomial-time guarantee?

1947 **Dantzig: simplex**

Change one basis at a time. This becomes a major practical method for solving LPs.

1979 **Khachiyan: ellipsoid method**

LP can be solved in time polynomial in the **binary input length**, including the bits used to encode rational coefficients.

1984 **Karmarkar: an interior-point breakthrough**

A new polynomial-time algorithm uses projective transformations and interior steps, opening a different algorithmic direction.

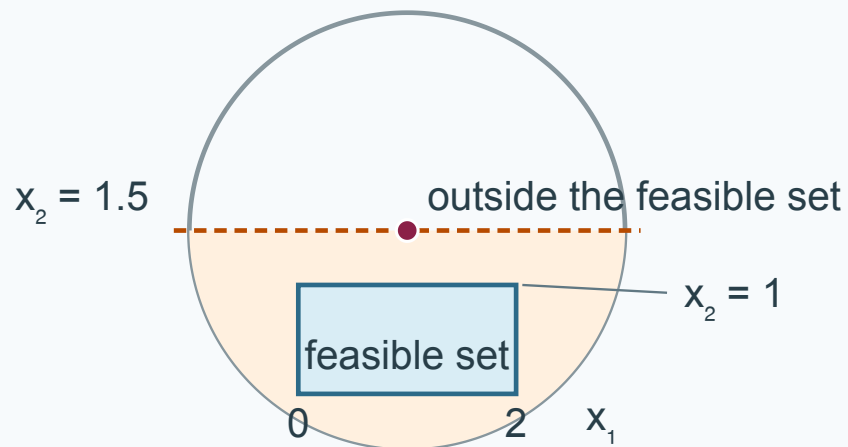
Study sources: [Dantzig's account](#) · [Khachiyan \(1979\)](#) · [Karmarkar \(1984\)](#).

Ellipsoid: shrink a region that still contains the answer

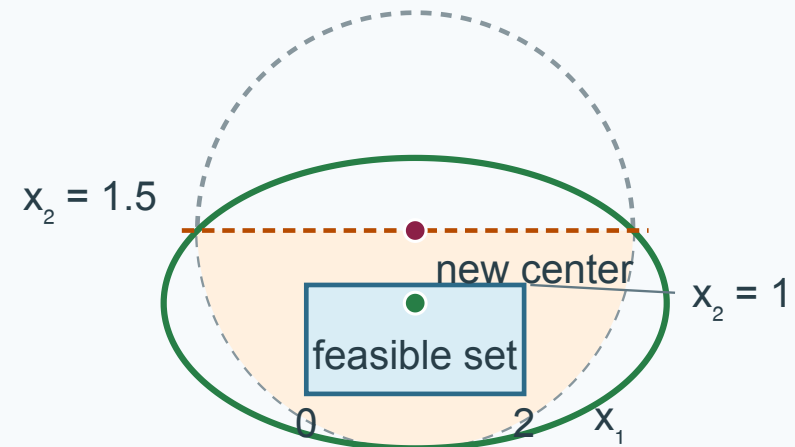
Same LP in both pictures: $\min f = -x_1 - x_2$, $0 \leq x_1 \leq 2$, $0 \leq x_2 \leq 1$.

The current center violates $x_2 \leq 1$. Keep the lower half through that center; every feasible point remains.

1. Cut through the current center



2. Enclose the retained half



Enclose the **entire retained half** in about 77% of the old area. Centers need not be feasible; at a feasible center, keep points with no worse objective.

Study sources: [Ellipsoid notes](#)

Interior points can approach the same optimum

Same LP in both pictures: $\min f = -x_1 - x_2, \quad 0 \leq x_1 \leq 2, \quad 0 \leq x_2 \leq 1.$

Orange: one improving simplex route follows two edges to the optimum.

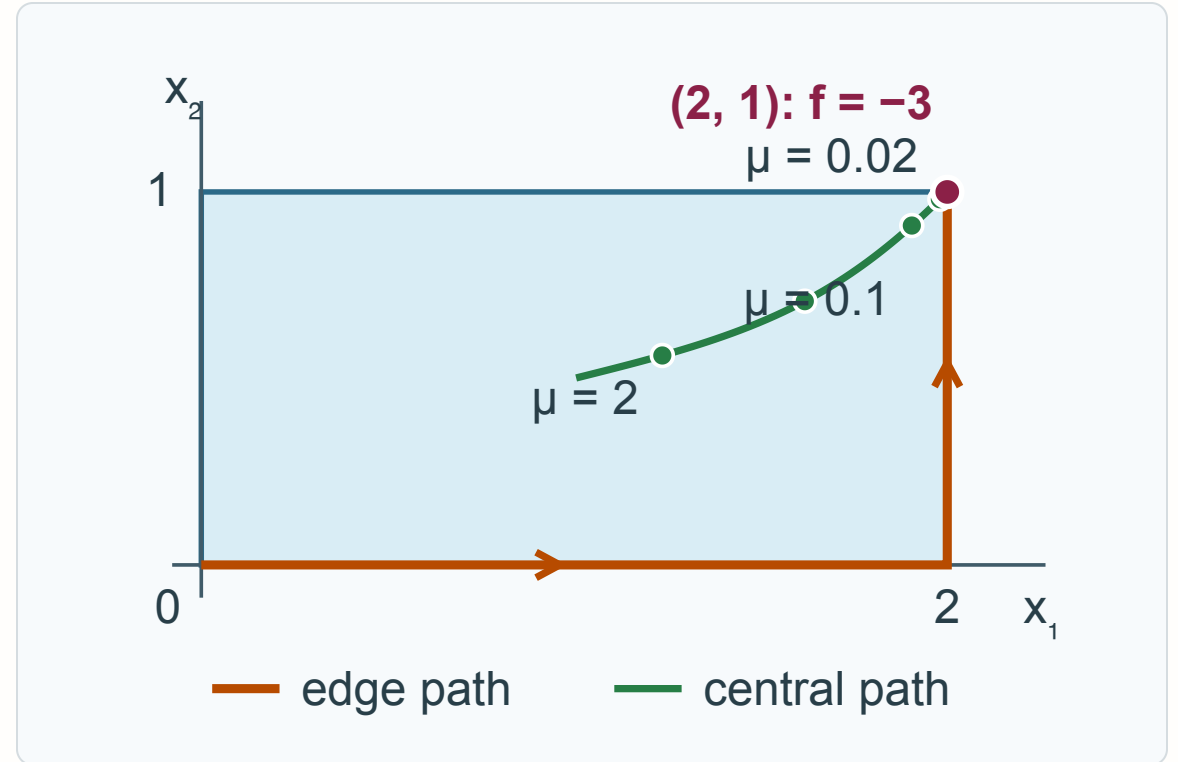
Green: a barrier penalizes approaching the boundary. For each $\mu > 0$, minimize cost plus this penalty.

As μ decreases, these **barrier minimizers** approach $(2, 1)$. Their curve is the **central path**.

These are exact barrier minimizers, not recorded algorithm steps. This illustrates the barrier approach, rather than Karmarkar's particular algorithm.

Barrier shown: $\Phi_\mu(x) = -x_1 - x_2 - \mu \log[x_1(2 - x_1)x_2(1 - x_2)]$.

Study sources: [Barrier methods and the central path](#)



Theory and practical speed answer different questions

Method	What it gives us	How to interpret it
Simplex	Basis changes and an optimal dictionary.	Widely used in practice; some pivot rules have exponential worst cases.
Ellipsoid	A polynomial-time guarantee for rational LPs.	A landmark in complexity theory; usually not the practical LP method of choice.
Interior-point	Polynomial-time methods and practical barrier solvers.	Modern solvers offer both simplex and barrier; which is faster depends on the problem.

Three separate ideas: avoiding cycling, bounding worst-case work, and solving a particular LP quickly.

Later in the course, we will study the interior-point idea conceptually. For now, keep these different goals separate.

Study sources: [Karmarkar \(1984\)](#) · [LP solver guide](#)

Practice: what does each statement guarantee?

1. A valid anticycling rule prevents repeated bases. Does that guarantee a short run?

No. It guarantees termination under the simplex assumptions, but the number of different bases can be large.

2. A cube has an exponentially long improving path. Must every rule follow it?

No. The construction supplies a possible slow path. Another entering choice can take a shortcut.

3. One tableau pivot costs $O(mn)$. Is total work now polynomial?

Not from this alone. We also need to bound the number of pivots.

Three ideas to carry into duality

A pivot changes a basis

In the column picture, a positive improving step lowers the attainable cost above the same target. Degeneracy can leave that cost unchanged.

Termination and speed are different questions

Work per pivot, number of pivots, and the shortest available path measure different things.

A bound can certify the answer

Our final supporting plane proved that no feasible mixture could cost less. Duality will turn this bounding idea into another LP.

Simplex and interior-point methods take different routes to an optimal solution. We will return to interior-point methods later in the course.

Study guide: connect a pivot to its picture and cost

Column geometry

Explain why mixing columns represents feasibility, why an improving pivot lowers cost, and how a supporting plane certifies the optimum. Reconstruct the bounded-set normalization in both directions, then use the dictionary and ratio test to explain the displayed pivot.

Computational work

Separate work per pivot from the number of pivots. Explain why finite termination and a short available path do not guarantee a fast chosen rule.

Algorithm history: explain the polynomial-time breakthrough and distinguish the ellipsoid enclosure from an interior-point path.

Textbook: Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, §§3.6–3.7.

Review: [Simplex III: Two-Phase Simplex](#) explains how to find a feasible starting basis.

Continue to optional proofs, worked Example 3.10, and implementation/complexity topics

Optional study: choose a topic

These sections extend the main lecture. Each keeps its definitions, calculations, and diagrams together.

Appendix A · Column geometry calculations

Appendix B · Example 3.10

Appendix C · Pivot work, paths, and proofs

Appendix D · Diameter and pivot paths

Appendix E · Average and smoothed analysis

A new picture: columns are points, variables are weights

Previously, a point represented x . Here, each column A_j is a point and x_j is its weight.

$$Ax = \sum_j x_j A_j = b.$$

For a matrix A with m rows, study this mixture-form LP:

$$\min c^\top x \quad \text{s.t.} \quad Ax = b, \quad \sum_j x_j = 1, \quad x \geq 0.$$

Nonnegative weights summing to one make b a **weighted average** of the columns. Each column lists coefficients in the equations.

The sum-to-one condition belongs to this model. Adding it to an arbitrary LP can change the feasible set.

We will use one five-column example to connect this picture to dictionary arithmetic.

One five-column LP for the whole picture

Choose weights x_B, x_C, x_D, x_E, x_F . The labels B–F identify the five columns:

Column	B	C	D	E	F
Equation 1: A_{1j}	1/2	3	0	2	0
Equation 2: A_{2j}	1/2	0	3	1/2	0
Unit cost: c_j	7	4	4	1	4

The target is $b = (1, 1)$. Column C contributes $3x_C$ to equation 1 and $0x_C$ to equation 2; its contribution to cost is $4x_C$.

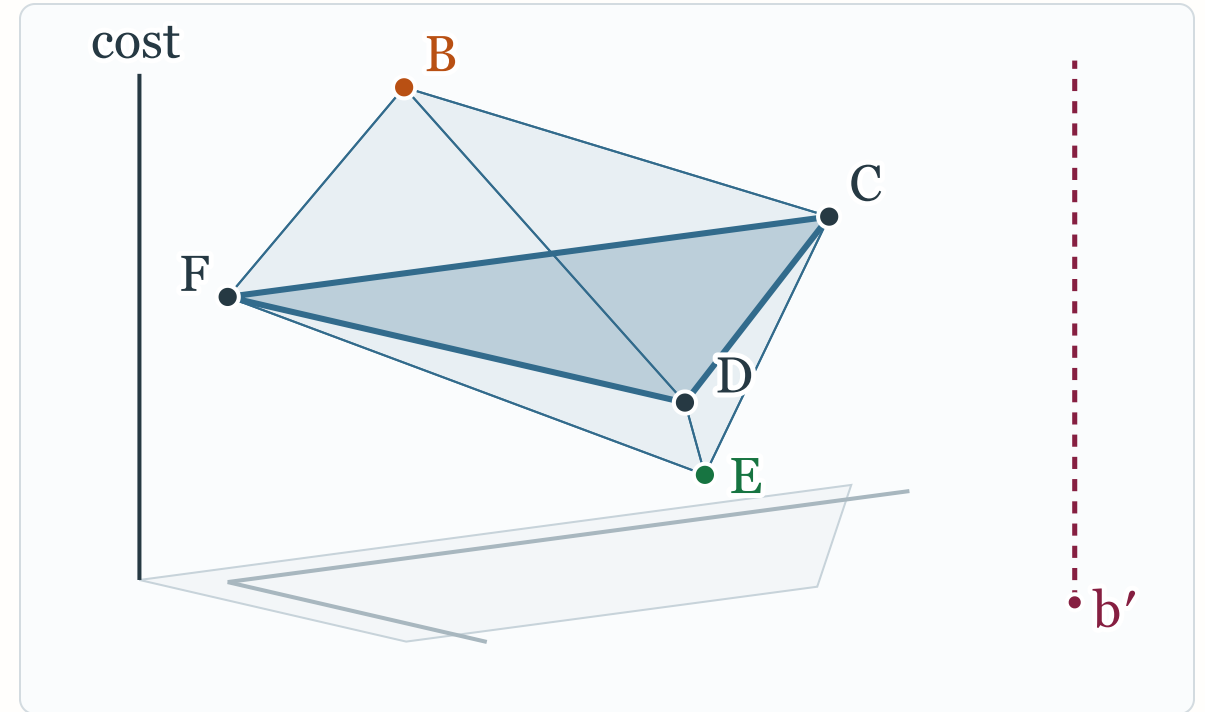
If the line misses the hull, no mixture is feasible

Keep the same five lifted columns, but change the requirement to $b' = (2.4, 2.4)$.

Every column satisfies $A_{1j} + A_{2j} \leq 3$.
Therefore every convex combination satisfies the same inequality.

$$b'_1 + b'_2 = 4.8 > 3.$$

The new requirement line never meets the hull.
The LP is infeasible, regardless of its objective.



A feasible basic mixture need not be optimal

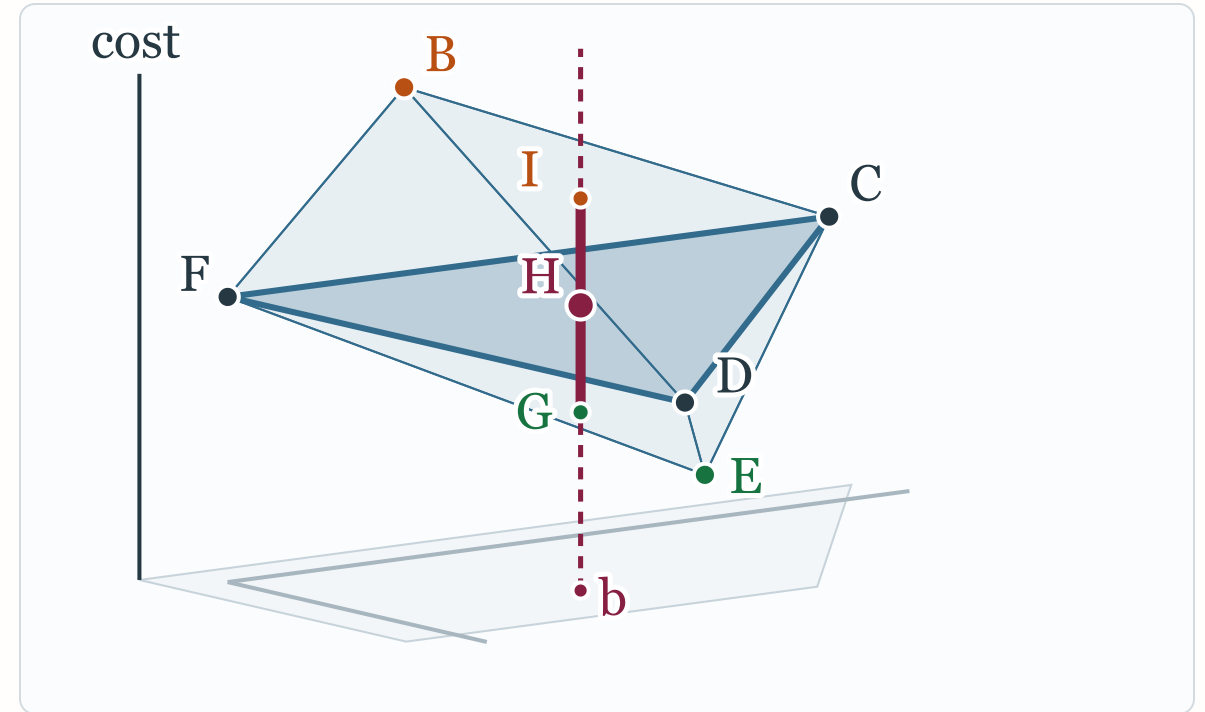
The same requirement line meets several possible basic triangles.

$$H = (1, 1, 4),$$

$$G = (1, 1, 5/2).$$

H uses CDF. The lower point G uses DEF. A change of basis can move the intersection downward while the requirement stays fixed.

The task is to choose a new basic point and an old basic point to remove.



The basic points lie on the plane $z = 4$

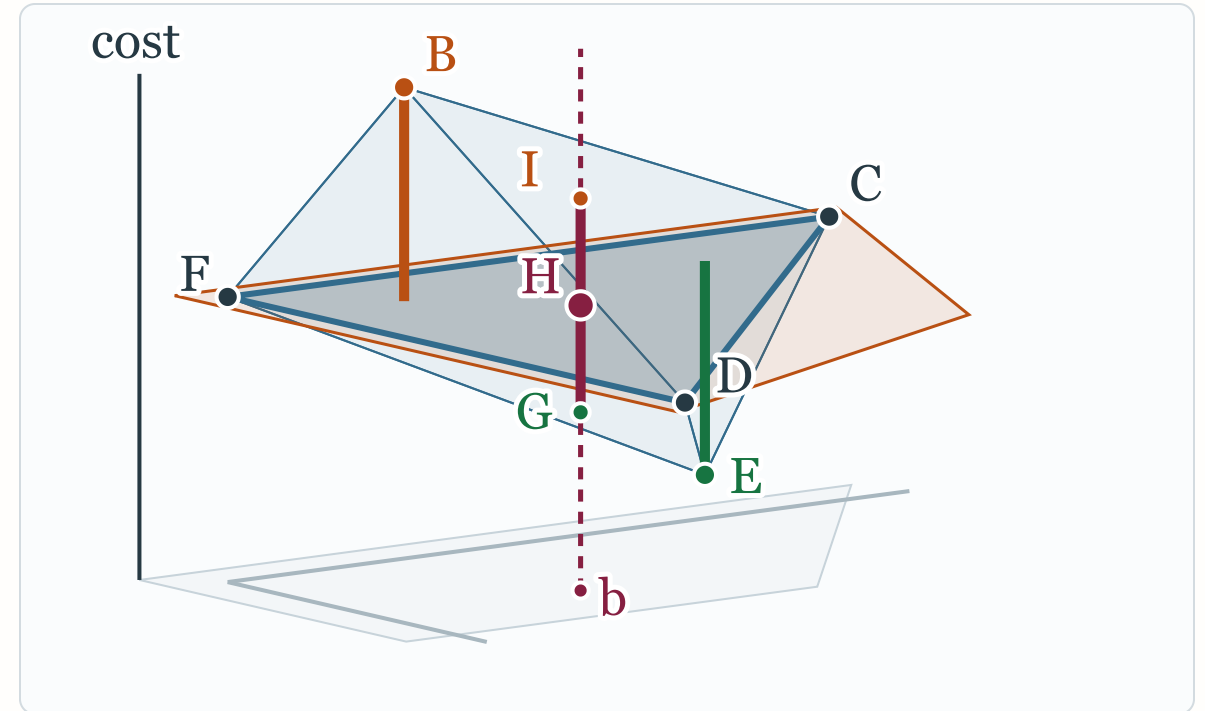
C, D, and F all have cost 4. Extend their triangle to its full plane:

$$z = 4.$$

E has height 1, below the plane. B has height 7, above it.

Compare this with the objective dictionary $z = 4 + 3x_B - 3x_E$. The coefficients match the signed height differences.

The next page makes that correspondence explicit.



Reduced cost is a signed vertical gap

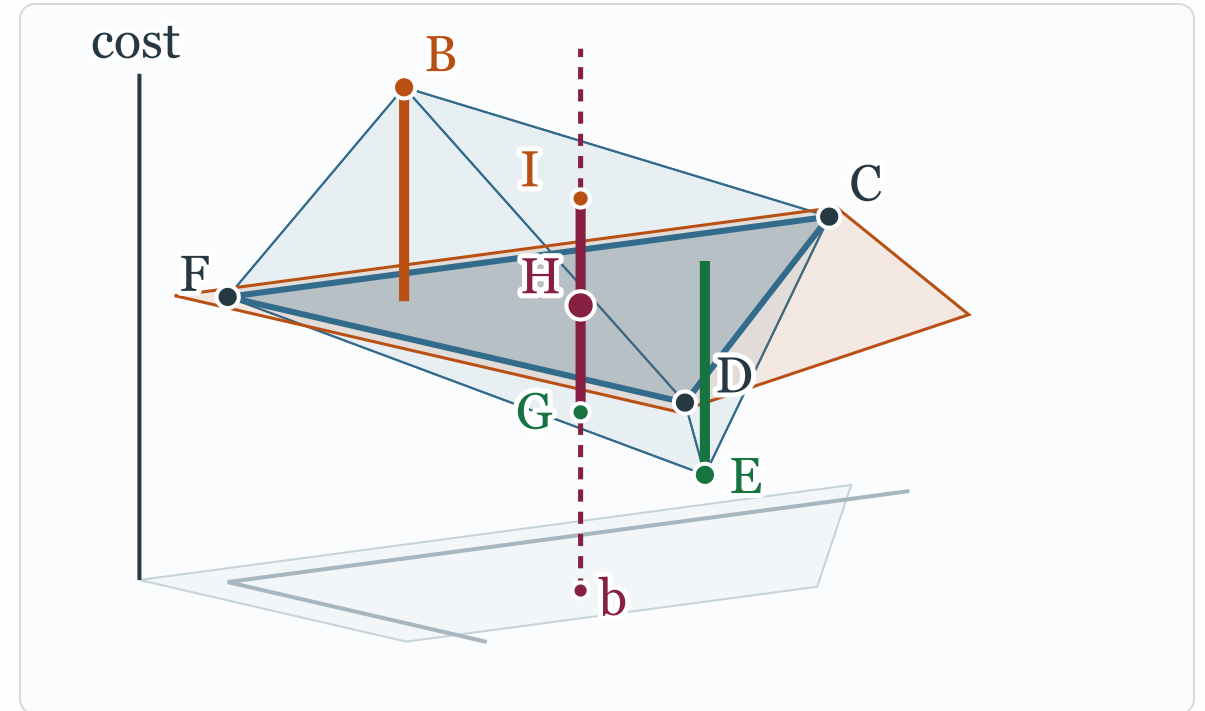
Subtract the height of the plane through CDF from the height of the entering point:

$$\begin{aligned}\bar{c}_E &= 1 - 4 = -3, \\ \bar{c}_B &= 7 - 4 = 3.\end{aligned}$$

$$z = 4 + 3x_B - 3x_E.$$

The same numbers are the objective coefficients after substitution.

A negative gap gives an improving slope.
Nonnegative basic weights still determine how far we can move.



B gives a feasible move, but raises the cost

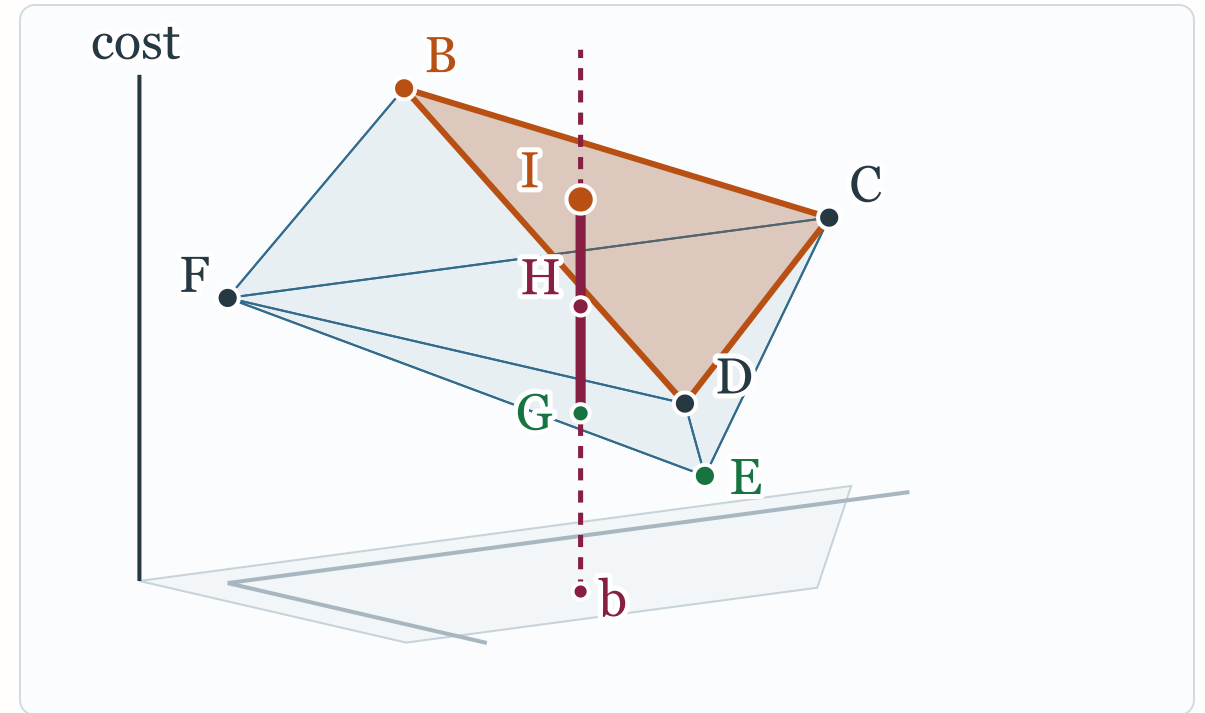
Keep $x_E = 0$ and increase x_B . The dictionary keeps every basic weight nonnegative until

$$x_B \leq \min\{2, 2, 1/2\} = 1/2.$$

F reaches zero first. The new triangle is BCD:

$$I = \frac{1}{2}B + \frac{1}{4}C + \frac{1}{4}D = (1, 1, 11/2).$$

The objective equation predicts $z = 4 + 3(1/2) = 11/2$. This feasible exchange worsens a minimization objective.



At the maximum step, E replaces C

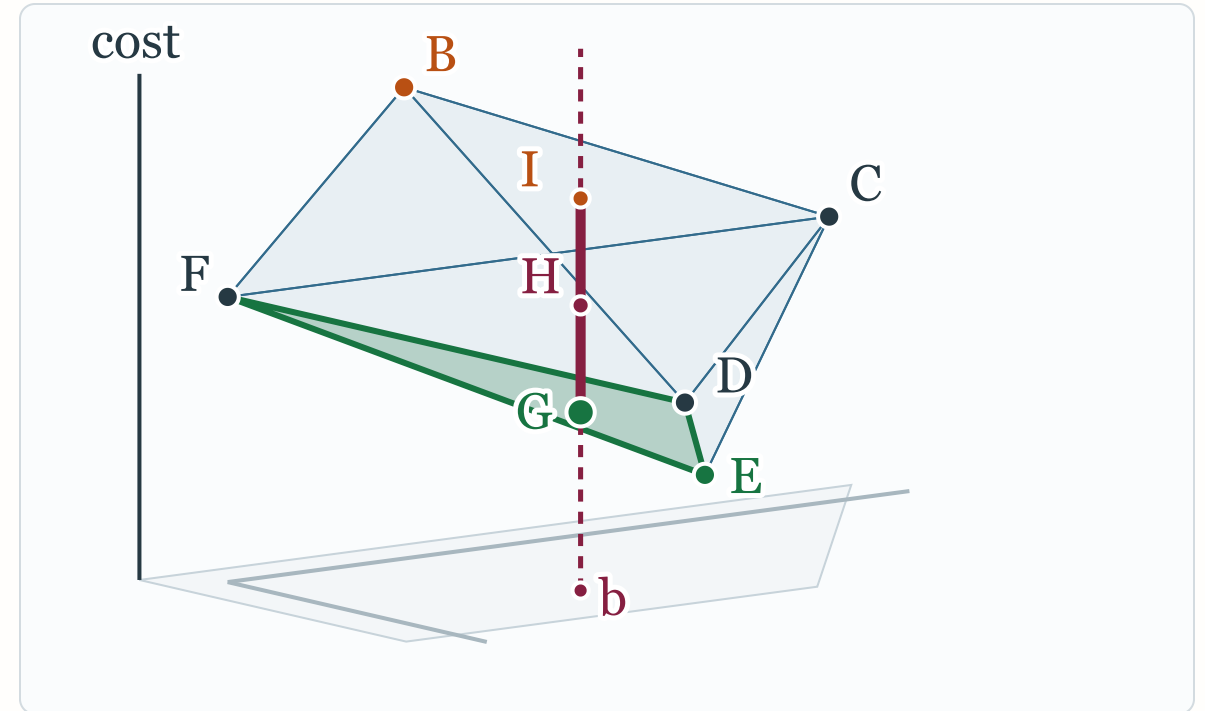
Use the ratio-test step $x_E = \theta^* = 1/2$, with $x_B = 0$. Substitution gives

$$x_C = 0, \quad x_D = x_F = \frac{1}{4}.$$

C leaves. The new basic weights use D, E, F:

$$G = \frac{1}{4}D + \frac{1}{2}E + \frac{1}{4}F = (1, 1, 5/2).$$

The dictionary predicts $z = 4 - 3(1/2) = 5/2$.
The new triangle meets the requirement line at exactly that height.



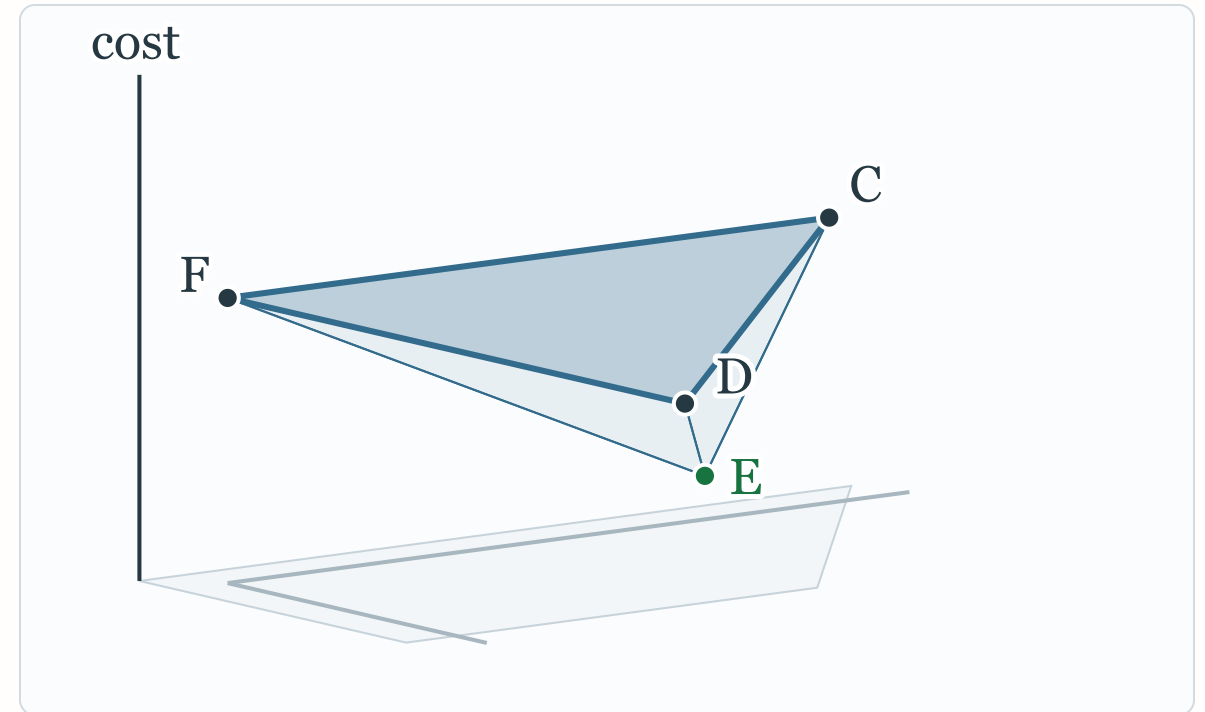
Four noncoplanar points give a tetrahedron

C, D, F form the old triangle. Adding E below that triangle creates a solid with four corners:

Its convex hull is a tetrahedron, also called a 3-simplex. It has four triangular faces.

C, D and F lie in one plane. E lies below it, so C, D, E and F are not coplanar.

The basis exchange selects a new boundary triangle of this solid.

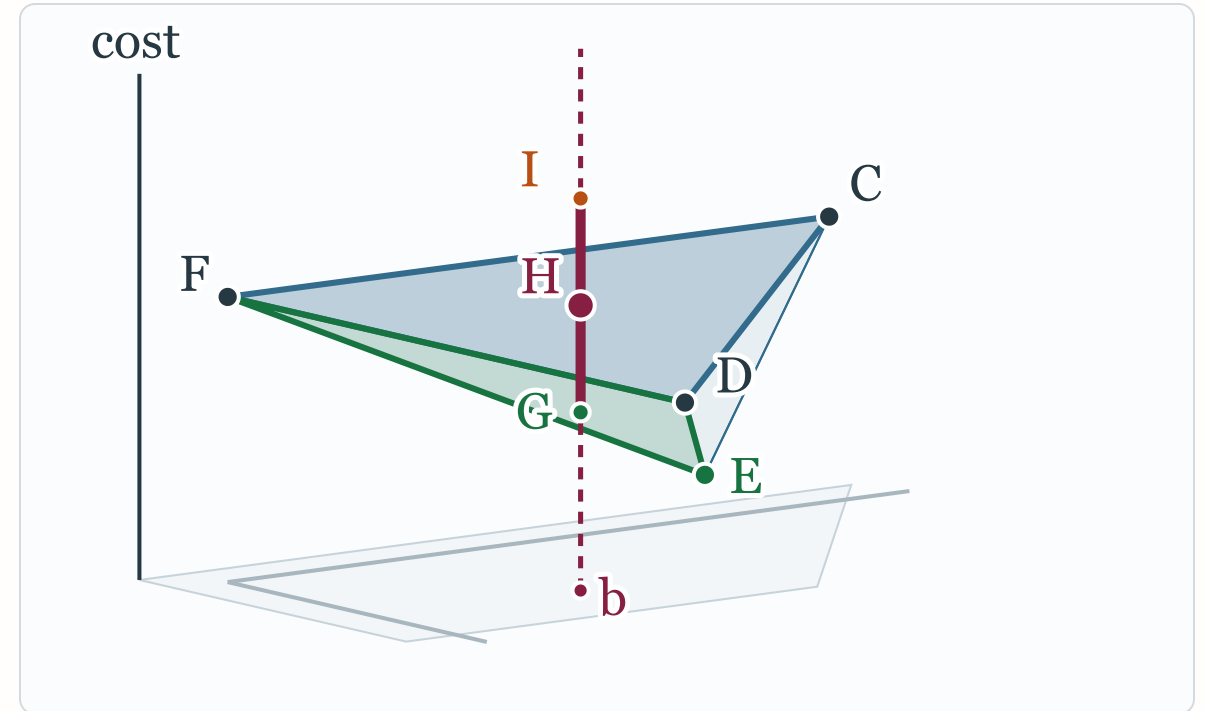


Adding E creates four possible boundary faces

The current triangle CDF together with E forms the tetrahedron CDEF.

Its four faces are CDF, CDE, CEF and DEF. The three new faces are candidates obtained by removing F, D or C.

In general an $(m + 1)$ -simplex has $m + 2$ faces: the old basic simplex plus $m + 1$ candidate replacements.

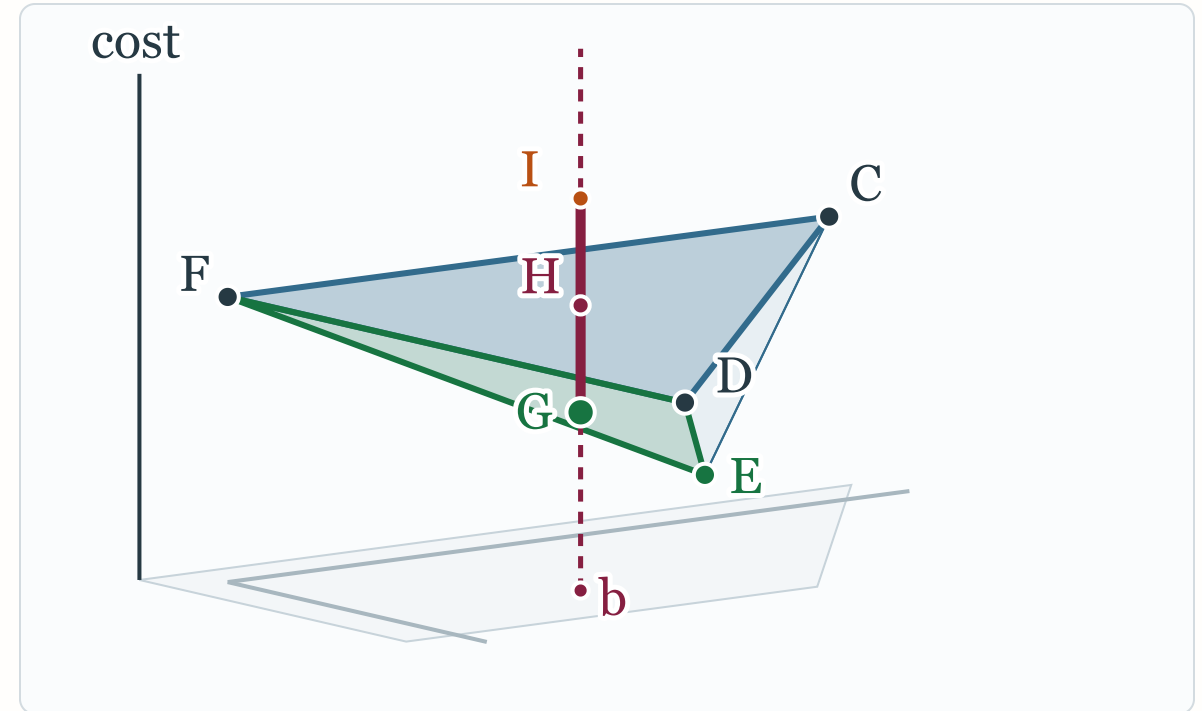


The requirement line selects the feasible new face

At H, the requirement line meets the old face CDF. Follow the line downward through the tetrahedron.

The next boundary intersection is G, on face DEF. That face contains E and excludes C: E enters and C leaves.

A candidate face that does not meet the requirement line cannot give nonnegative weights for this requirement.



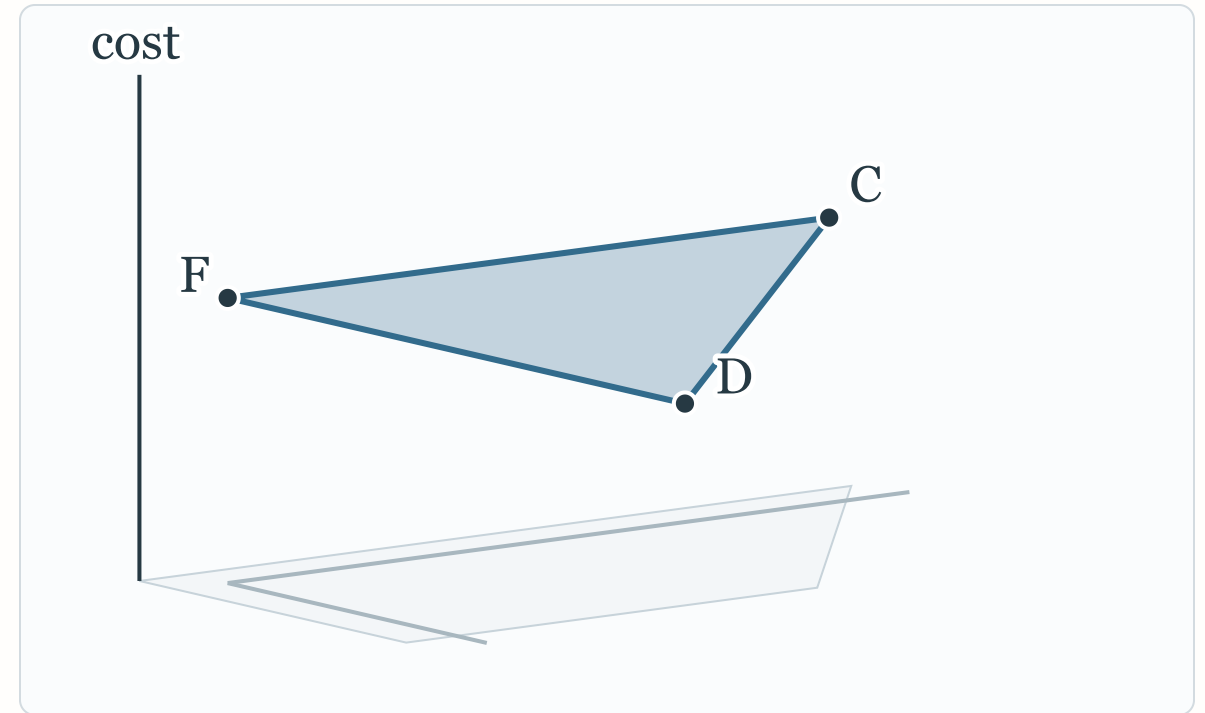
Affine independence gives a nonflat simplex

Points $v_1, \dots, v_{k+1} \in \mathbb{R}^n$ are **affinely independent** when

$$v_1 - v_{k+1}, \dots, v_k - v_{k+1}$$

are linearly independent. Necessarily $k \leq n$.

Their convex hull is a **k -dimensional simplex**.
Three noncollinear points give a triangle: CDF is a 2-simplex.



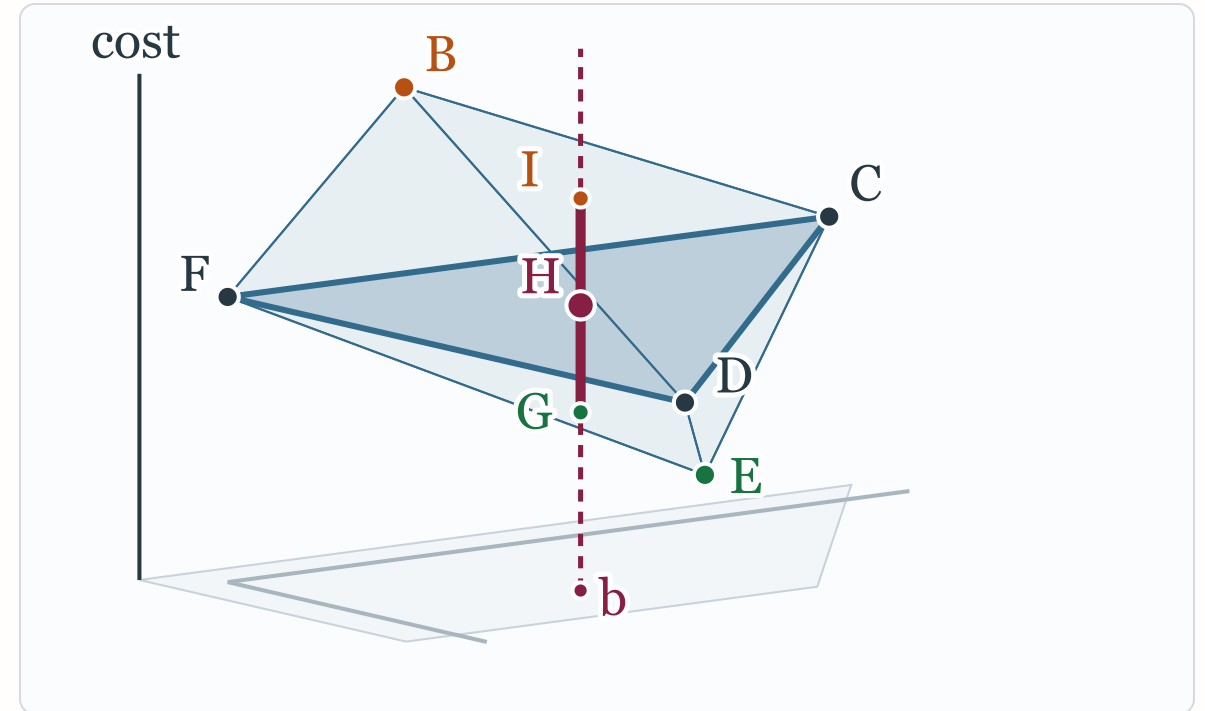
A basis selects three lifted points

Including convexity, the constraint matrix is

$$\tilde{A} = \begin{pmatrix} A \\ \mathbf{1}^\top \end{pmatrix}, \quad \tilde{b} = \begin{pmatrix} b \\ 1 \end{pmatrix}.$$

Assume $\tilde{A}$ has full row rank. A basis selects $m + 1$ independent columns $(A_j, 1)$.

Here $m = 2$: choose C, D, F. Their lifted versions are the **basic points**. B and E are nonbasic.

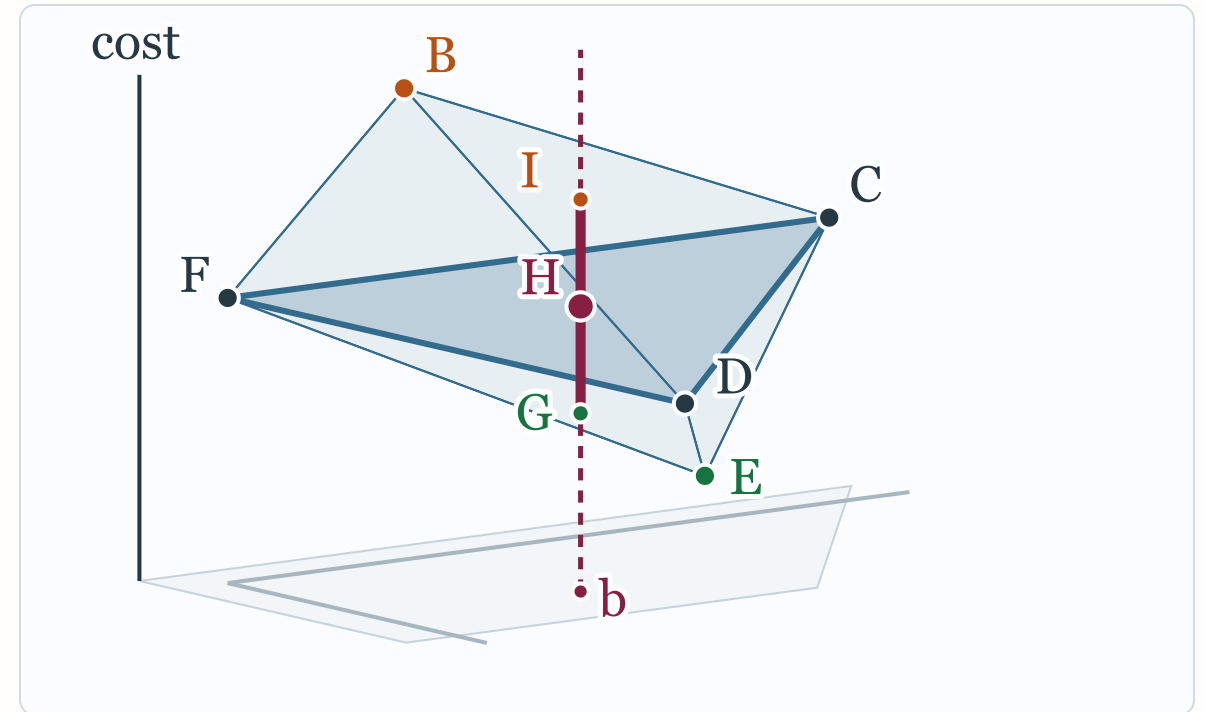


The selected points form the basic simplex

The basic columns $(A_j, 1)$ are independent, so their horizontal points A_j are affinely independent.

Lifting them to (A_j, c_j) preserves that affine independence: a relation between lifted differences would also be a relation between horizontal differences.

Their convex hull is the basic simplex. Our three basic points form the shaded triangle CDF.



The general plane formula recovers reduced costs

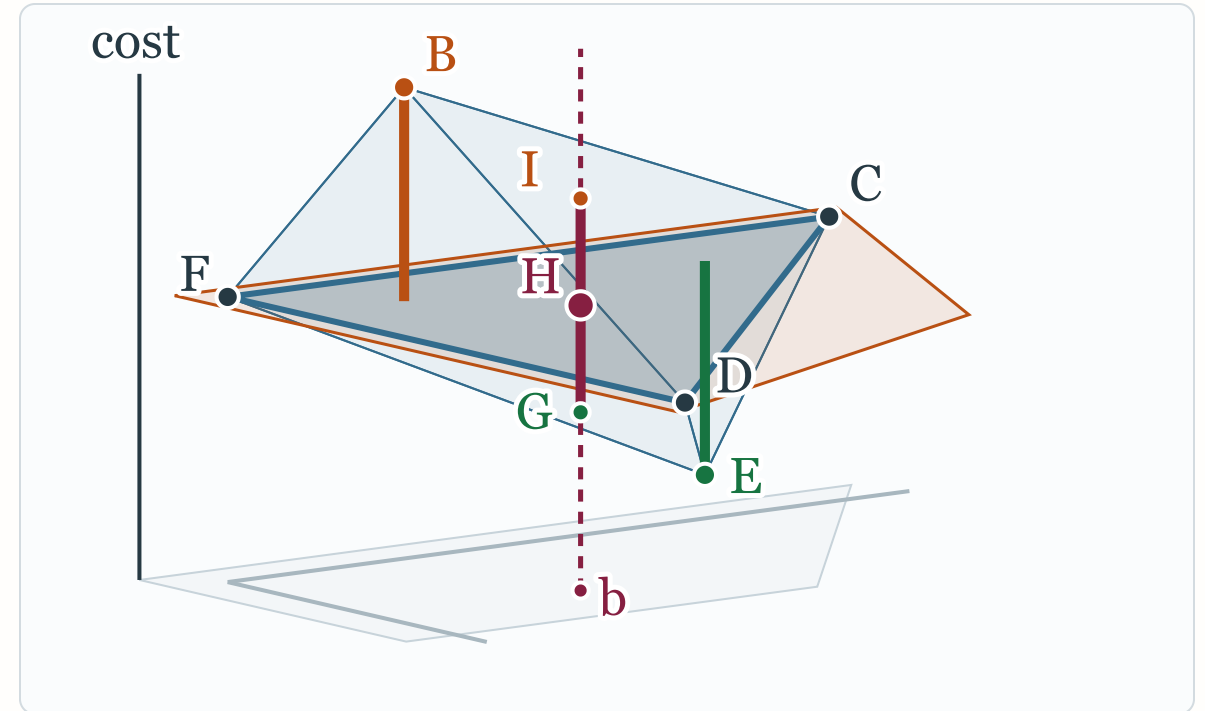
Write the plane through the basic points as $z = \pi^\top a + \alpha$. Matching their costs gives

$$\widetilde{B}^\top \begin{pmatrix} \pi \\ \alpha \end{pmatrix} = c_B.$$

$$\bar{c}_j = c_j - (\pi^\top A_j + \alpha).$$

Subtract plane height from point height. This equals the coefficient obtained by dictionary substitution.

For CDF, $\pi = (0, 0)$, $\alpha = 4$. Duality will interpret these plane coefficients as constraint weights.



Example 3.10: eight points, one horizontal coordinate

With $m = 1$, the basic simplex is a line segment. Use the following eight points:

j	1	2	3	4	5	6	7	8
A_j	5	1	-2	-3	3	4	2	-1
c_j	7	6	6	5	2	8	5	3

$$\min \sum_{j=1}^8 c_j x_j \quad \text{s.t.} \quad \sum_{j=1}^8 A_j x_j = 0, \quad \sum_{j=1}^8 x_j = 1, \quad x \geq 0.$$

We will choose the point with the most negative reduced cost: the largest downward vertical gap from the line through the basic points.

Initial basis 3, 6: point 5 has the largest downward gap

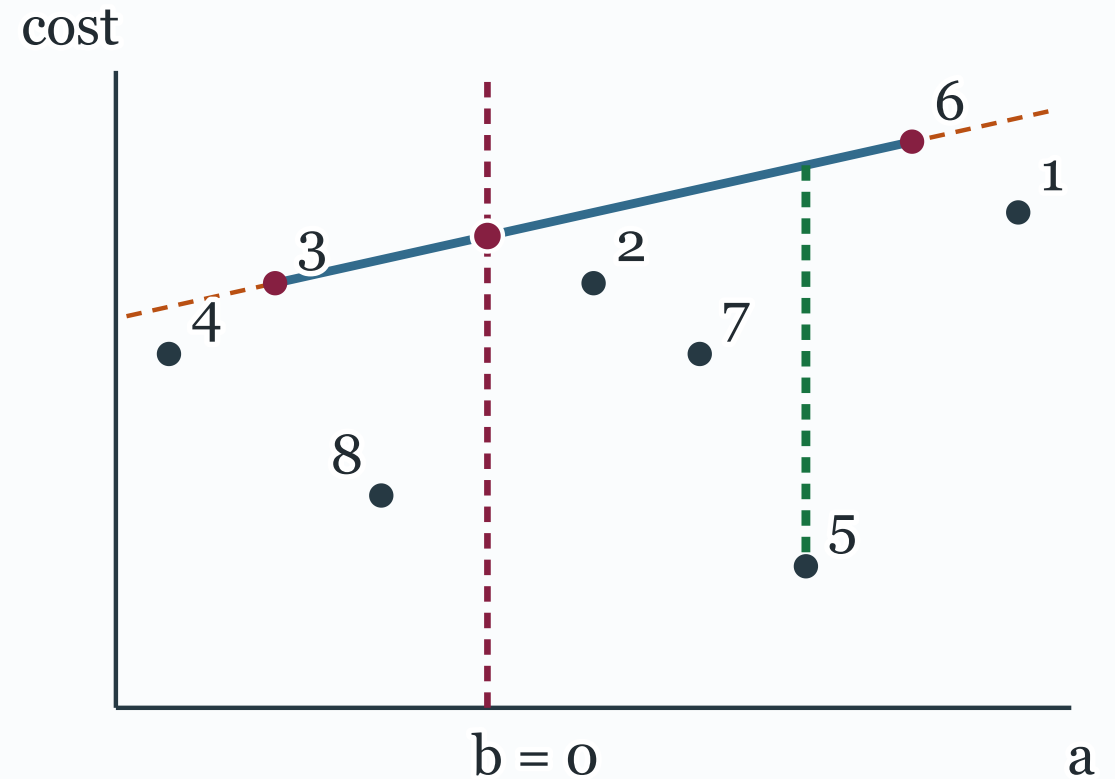
The line segment between points 3 and 6 crosses the requirement line.

$$x_3 = \frac{2}{3}, \quad x_6 = \frac{1}{3}, \quad z = \frac{20}{3}.$$

$$z = \frac{1}{3}a + \frac{20}{3}.$$

Point 5 has gap $2 - 23/3 = -17/3$, the most negative reduced cost.

Point 5 enters. The requirement lies between 3 and 5, so point 6 leaves.



Next basis 3, 5: point 8 is now the best candidate

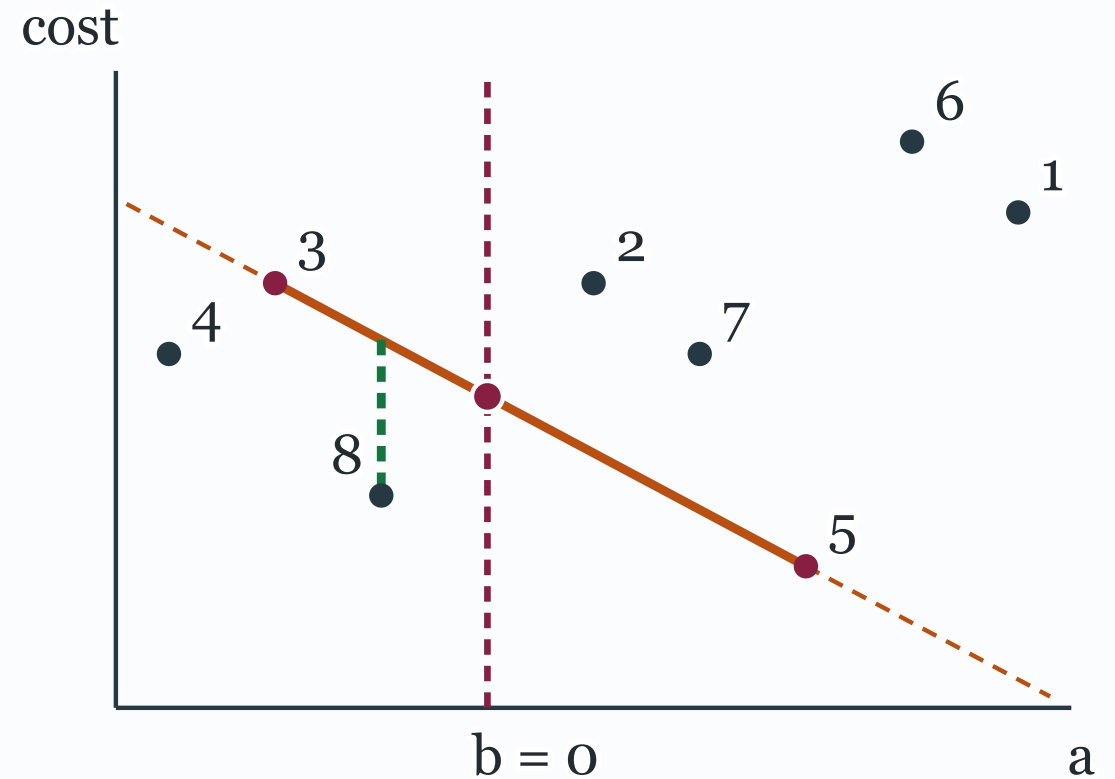
After the first pivot, the segment is 3, 5.

$$x_3 = \frac{3}{5}, \quad x_5 = \frac{2}{5}, \quad z = \frac{22}{5}.$$

$$z = -\frac{4}{5}a + \frac{22}{5}.$$

Now point 8 has the most negative gap: $3 - 26/5 = -11/5$.

Point 8 enters and point 3 leaves. The new segment must still cross the requirement line.



Final basis 5, 8: every point is on or above the line

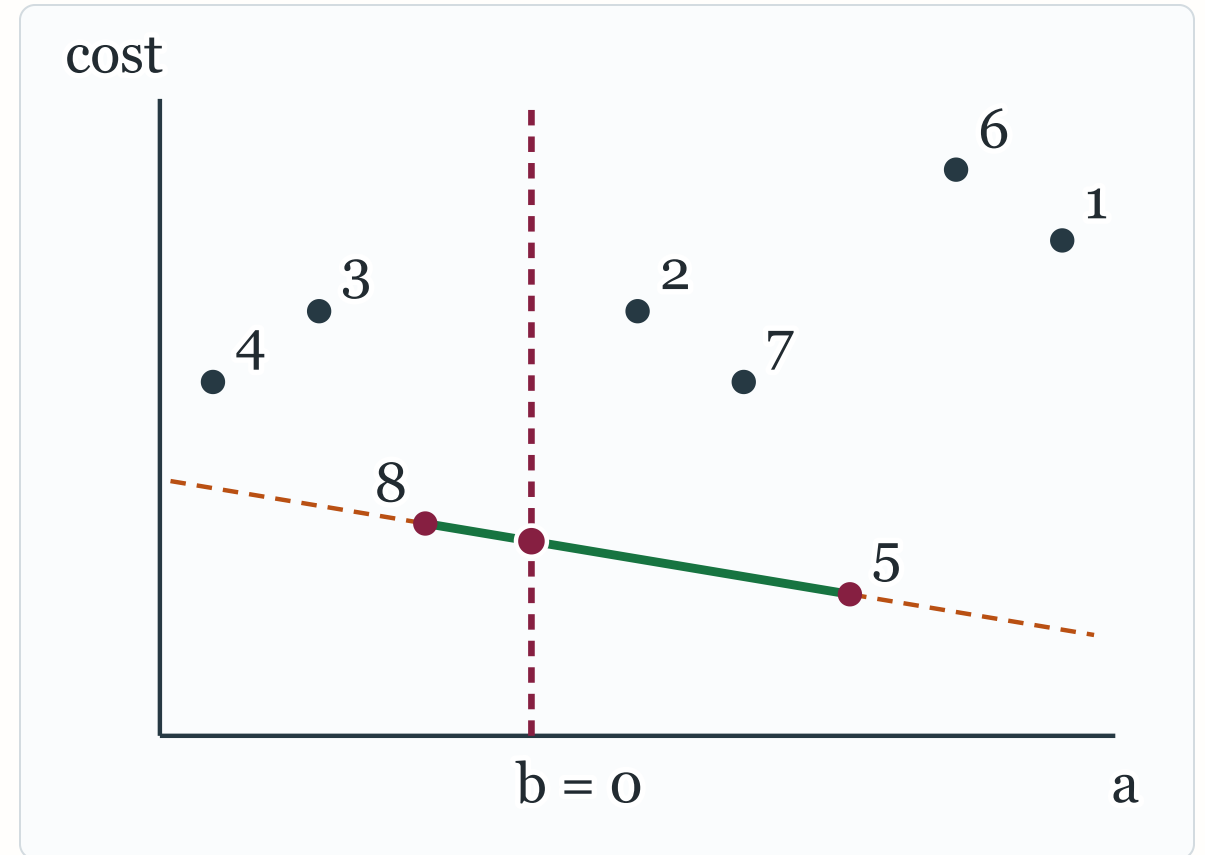
The final segment uses points 5 and 8.

$$x_5 = \frac{1}{4}, \quad x_8 = \frac{3}{4}, \quad z = \frac{11}{4}.$$

$$z = -\frac{1}{4}a + \frac{11}{4}.$$

All remaining points are above this line. Every nonbasic reduced cost is positive.

The current feasible mixture attains the lowest possible height at the requirement.



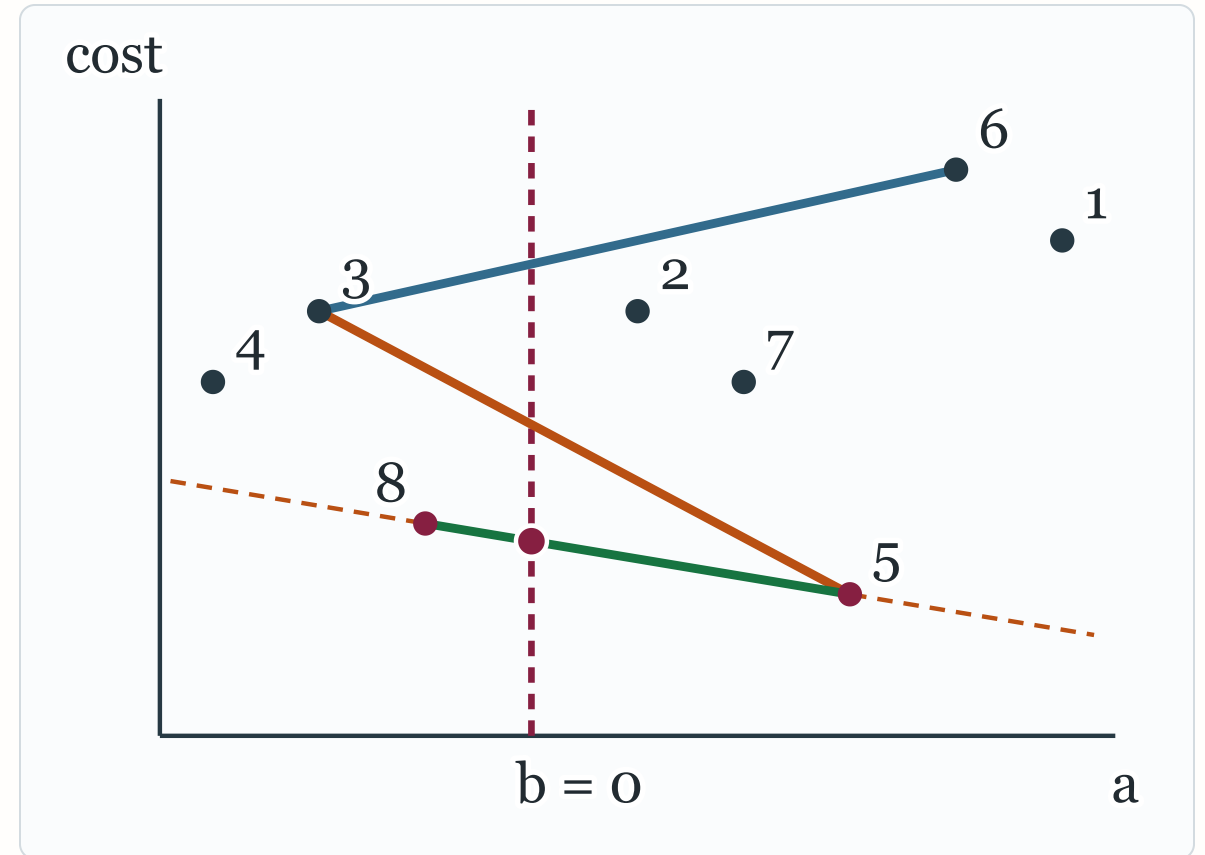
Read the entire path: $3, 6 \rightarrow 3, 5 \rightarrow 5, 8$

Each pivot changes one endpoint of the basic segment while preserving its intersection with the requirement line.

$$\frac{20}{3} > \frac{22}{5} > \frac{11}{4}.$$

Self-study: recover the weights by solving the two equalities for each pair. Then calculate every reduced cost to confirm the entering choices.

The 2D segment rotates at a shared endpoint. The 3D triangle pivots around a shared edge. Both pictures describe the same basis exchange.



— initial: 3, 6 — next: 3, 5 — optimal: 5, 8

First factor: update the entries in one tableau

With m independent equations and n variables, a tableau has roughly mn coefficient entries.

A pivot rescales one row, then subtracts multiples of it from the other rows. Each entry needs a bounded number of arithmetic operations.

$$\text{work per tableau pivot} = O(mn).$$

Here $O(mn)$ means that the work grows at most proportionally to mn , up to a constant factor.

Revised simplex stores information for computing the needed rows and columns instead of updating a complete tableau. It represents the same mathematical pivot.

Neither representation tells us how many pivots will be needed.

Second factor: how many pivots?

In practice, many LPs need relatively few pivots. Observed counts can grow roughly with the dimensions, but this is an empirical observation.

That observation is not a bound for every LP. The feasible region can have exponentially many vertices.

To show slow behavior, we need more than many vertices: an **improving edge path** that a pivoting rule actually follows.

A cube already has exponentially many corners

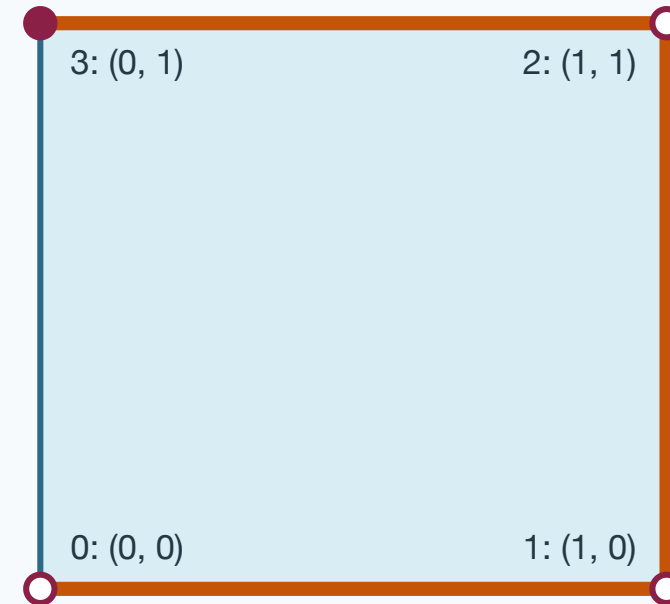
Use the objective and unit cube

$$\min f = -x_n, \quad 0 \leq x_i \leq 1.$$

There are n coordinates, $i = 1, \dots, n$.

At a vertex each coordinate is either 0 or 1. There are 2^n binary choices.

A spanning path visits every vertex exactly once using edges.



minimize $f = -x_2$

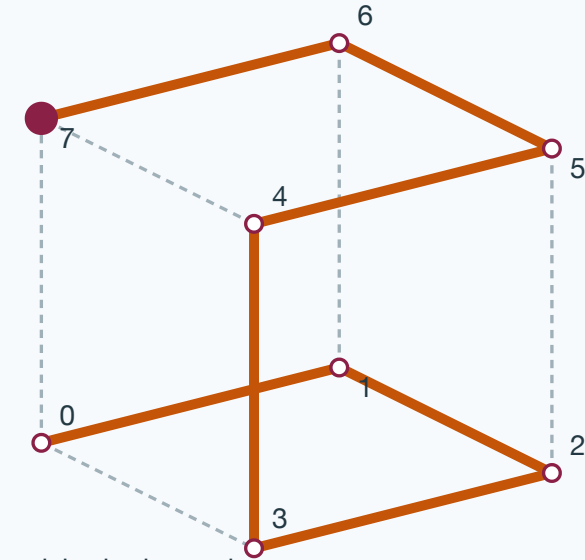
In two dimensions: 4 vertices and a path of 3 edges.

Build the path one dimension at a time

1. Follow the square path on $x_3 = 0$.
2. Cross one edge to $x_3 = 1$.
3. Follow the same square path in reverse.

$$L_n = 2L_{n-1} + 1 = 2^n - 1.$$

The same construction works recursively in every dimension.



Vertex 7 of 7; $x = (0, 0, 1)$; $f = -1$.

Why the ordinary cube is not enough

Recall the minimization objective $f(x) = -x_n$.

On the cube, half the vertices have $f = 0$, and half have $f = -1$.

Our spanning path has many edges along which the objective stays constant. It does not yet demonstrate a long sequence of *strictly improving* pivots.

Next we tilt the bounds, so movement in an earlier coordinate changes the possible value of the last coordinate.

A bad path does not settle every pivoting rule

The tilted cube gives a long improving path, but it also has a one-pivot shortcut.

Separate the claims: a bad example for one rule does not prove that every possible rule must be slow.

For many particular rules, exponential examples are known. Finding a general polynomial-time pivot rule is a different question.

Disser and Mosis (2025) describe that general question as open. Their all-rule lower bound concerns a nonlinear active-set problem, not linear programming.

Worst-case behavior does not describe every instance

A worst-case result asks how badly a method can behave. An average-case result asks how it behaves under a specified distribution of inputs.

Many practical LPs need modest pivot counts, sometimes growing roughly linearly with problem dimensions. This is an observation, not a universal guarantee.

To make “usually fast” precise, specify a probability model. The optional study section distinguishes average-case and smoothed analysis.

Practical choices also include using sparse data and comparing simplex with interior-point methods, which we will study later.

Practice: identify the outcome from a dictionary

All variables are nonnegative. A minimization LP has this feasible dictionary, with x_1 nonbasic:

$$\begin{aligned}f &= 3 - x_1, \\x_2 &= 1 + x_1, \\x_3 &= 2 + 2x_1.\end{aligned}$$

Can x_1 enter? Which variable leaves, and what does this tell us?

Answer: $x_1 = t \geq 0$ is feasible for every t : both basic variables increase. No variable limits the step.

$$(x_1, x_2, x_3) = (t, 1 + t, 2 + 2t), \quad f = 3 - t \rightarrow -\infty.$$

The LP is unbounded below. Detecting this ray finishes the algorithm; it does not require infinitely many pivots.

Implementation detail: pricing, factors, and arithmetic

For standard form, let m be the number of independent equations and n the number of variables.

A full tableau has $O(mn)$ entries. One pivot updates them by row operations: $O(mn)$ arithmetic operations.

Revised simplex maintains and updates basis factors. Dense worst-case pricing and updates can also cost $O(mn)$, since $m \leq n$.

A fresh dense factorization costs $O(m^3)$. Sparsity matters; arithmetic counts also omit the bit length of exact numbers.

Bland's entering scan: compute reduced costs in increasing variable-index order; stop at the first negative one. If none is negative, check all nonbasic variables.

Why Lexicographic Comparison Needs More Than a Ratio

For basis B , write $\bar{b} = B^{-1}b$ and $\bar{A} = B^{-1}A$. Suppose x_j enters; consider a row with $\bar{a}_{ij} > 0$.

Ordinary test: compare one ratio. **Lex rule:** if those ratios tie, compare further entries in the fixed column order.

$$\frac{\bar{b}_i}{\bar{a}_{ij}} \quad \text{versus} \quad \frac{1}{\bar{a}_{ij}} \left[\bar{b}_i \mid \bar{a}_{i1}, \dots, \bar{a}_{in} \right].$$

Full tableau: store $[\bar{b} \mid \bar{A}]$. Read each candidate row and divide it by its positive entering-column coefficient.

Revised simplex: store basis factors for solving linear systems, instead of the full tableau. Obtain the needed normalized row for lex comparison by linear solves, without explicitly forming the inverse.

Why there are exactly 2^n vertices

Choose a lower or upper bound for each coordinate, starting with x_1 .

The resulting triangular system has diagonal coefficients 1, so it defines a unique point.

Each lower bound is strictly below its upper bound: $\epsilon x_{i-1} < 1 - \epsilon x_{i-1}$, since $x_{i-1} \leq 1$ and $\epsilon < \frac{1}{2}$.

At most one bound per coordinate can be tight. A vertex needs n independent tight bounds, hence exactly one per coordinate. Every binary choice gives one distinct vertex.

Why reversing the upper path preserves improvement

On the lower face, $x_n = \epsilon x_{n-1}$: increasing x_{n-1} increases x_n .

Cross to the upper face, $x_n = 1 - \epsilon x_{n-1}$. This increases x_n , because $\epsilon x_{n-1} < 1 - \epsilon x_{n-1}$.

Then traverse the previous path backward. Now x_{n-1} decreases, so $1 - \epsilon x_{n-1}$ increases.

Consecutive vertices share $n - 1$ independent tight bounds. They are adjacent, and $-x_n$ strictly decreases throughout.

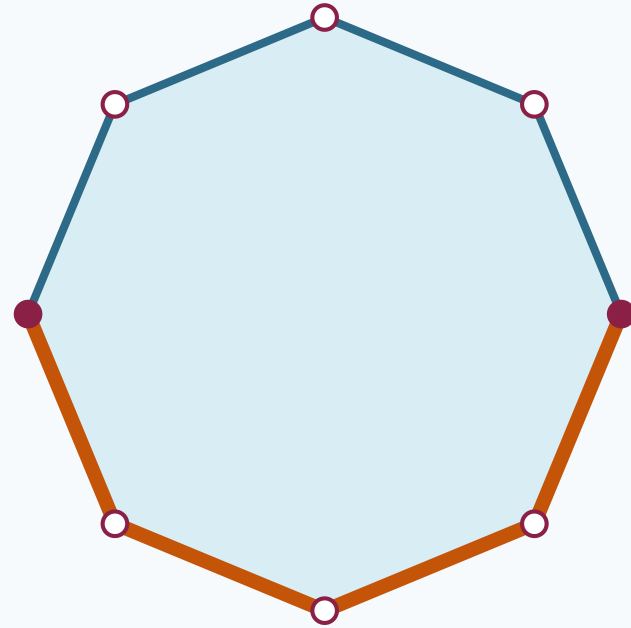
In two dimensions, a bounded polygon is a cycle

Here m counts inequality constraints and n is the ambient dimension. A bounded full-dimensional polygon has at most m edges.

Between two vertices, choose the shorter route around the boundary.

$$D(P) \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

The illustrated octagon attains 4.



cycle: $D = 4 = \text{floor}(m / 2) \cdot m = 8$

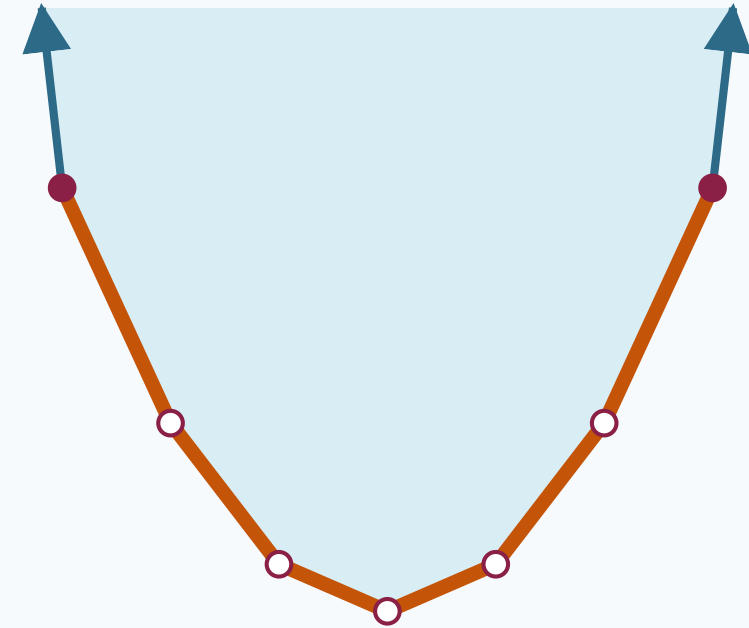
An unbounded polygon can have a longer chain

For a pointed, full-dimensional unbounded polygon, two boundary edges are rays. The finite vertices form a chain.

$$D(P) \leq m - 2.$$

With $m = 8$, the illustrated chain has seven vertices and six finite edges.

Thus the worst case over bounded and unbounded polygons is $\Delta(2, m) = m - 2$ for $m \geq 3$.



chain: $D = 6 = m - 2 \cdot m = 8$

Why diameter constrains every edge-following method

Suppose $d(v, w) = D(P)$. Start at v , and choose an objective for which w is the unique optimal vertex.

Every edge-following route from v to w needs at least $D(P)$ edge moves.

Define $\Delta(n, m)$ as the largest diameter among pointed n -dimensional polyhedra described by m inequalities.

A polynomial bound on diameter is necessary for a universally short edge path. It would not by itself tell us how to choose the pivots.

A weaker question asks for a polynomial bound

The **polynomial Hirsch conjecture** asks whether $\Delta(n, m)$ is bounded by some polynomial in n and m .

A classical bound is

$$\Delta(n, m) \leq m^{1+\log_2 n} = (2n)^{\log_2 m}.$$

The exponent depends on the dimension. This is a quasipolynomial bound, not a polynomial bound jointly in both parameters.

The false linear conjecture and the weaker polynomial question must be kept separate.

A short path need not give a fast algorithm

Suppose a polynomial diameter bound were proved. Would that make simplex a polynomial-time algorithm?

No. Diameter promises some short edge path. It does not promise an improving path, identify the path, or bound the work needed to find it.

Degenerate pivots may also change the basis without traversing an edge to a new vertex. Their number needs its own control.

Define the random experiment before taking an average

1. Fix the problem dimensions and choose a probability distribution over LP data.
2. Specify the algorithm, its initialization, and its pivoting rule.
3. Let T count the iterations and study $\mathbb{E}[T]$.

There is no single natural uniform distribution over all LPs. A result for one random model need not describe a different application.

One model: randomly orient fixed inequalities

Fix $c, a_1, \dots, a_m \in \mathbb{R}^n$ and $b_1, \dots, b_m$. Independently choose, with equal probability,

$$a_i^\top x \leq b_i \quad \text{or} \quad a_i^\top x \geq b_i.$$

There are 2^m sign choices. Suppose $L > 0$ produce feasible LPs.

For this model, Haimovich (1983) bounds the average by $n/2$ pivots over those L feasible problems, for the specially specified pivoting procedure.

The distribution and procedure are essential assumptions; this is not a bound for arbitrary simplex implementations.

Smoothed analysis: perturb an arbitrary starting instance

Average-case analysis randomizes the whole input. **Smoothed analysis** starts with arbitrary data and adds small random perturbations.

For an appropriate shadow-vertex simplex algorithm, Gaussian perturbation models give polynomial expected complexity in the problem dimensions and $1/\sigma$, where σ measures the noise scale.

This helps explain why carefully constructed difficult instances can be fragile. It does not give every pivoting rule a polynomial worst-case guarantee.

Spielman and Teng, Smoothed Analysis (2004).